

30-DAY BFS

FULL-STACK + AI MASTERY PLAN

360 hours | 12 study hours/day | 8:00 AM - 10:00 PM | Two 1-hour breaks

BFS / spiral learning: broad first pass -> deeper second pass -> production integration -> interview simulation

Primary identity	Senior Java / Spring Boot backend engineer with strong full-stack, cloud, distributed-systems and AI capability
Daily cadence	8-10 DSA 10-12 Java/Spring 12-1 break 1-3 System Design 3-5 Data 5-6 break 6-8 Frontend 8-9 DevOps/AWS 9-10 AI
Learning rule	Every block ends with code, diagram, interview answers, tests, or a measurable artifact.
Project laboratory	Use PAT-FOS as the recurring system: Spring Boot + Kafka + PostgreSQL + Redis + MongoDB + React + Docker.

How to use this guide

- Follow the time blocks, not the video completion percentage. You are cherry-picking high-value 2026 course sections because you already have 4-5 years of Java/Spring experience.
- For Java/Spring blocks, default to 30-45 minutes of targeted course material and 75-90 minutes of active practice/interview explanation. For newer areas (AI/Kubernetes), course time can be higher.
- Udemy curricula can be renamed or reordered. Navigate by the section title and lecture/topic title shown here rather than relying on a numeric lecture number.
- If a block is already GREEN (4-5/5), shorten video time and spend the saved time on implementation, failure analysis, or interview follow-ups.
- If a block is RED (1-2/5), record the exact miss and revisit it in BFS Pass 2. Do not derail the whole day by spending six hours on one topic.
- Breaks are mandatory: 12:00-1:00 PM and 5:00-6:00 PM. Step away from the screen.

2026 Course Reference Stack

Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

<https://www.udemy.com/course/java-in-depth-become-a-complete-java-engineer/>

Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

<https://www.udemy.com/course/spring-hibernate-tutorial/>

Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

<https://www.udemy.com/course/system-design-interview-prep/>

SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

<https://www.udemy.com/course/sql-and-postgresql/>

Redis: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; Redis 7; 22 sections / 187 lectures

<https://www.udemy.com/course/redis-the-complete-developers-guide-p/>

MongoDB - The Complete Developer's Guide - Maximilian Schwarzmuller / Academind

Updated Jan 2026; 19 sections / 266 lectures

<https://www.udemy.com/course/mongodb-the-complete-developers-guide/>

Mastering TypeScript - 2026 Edition - Colt Steele

Updated Jun 2026; 19 sections / 151 lectures

<https://www.udemy.com/course/learn-typescript/>

React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

<https://www.udemy.com/course/react-the-complete-guide-incl-redux/>

Docker & Kubernetes: The Practical Guide - Maximilian Schwarzmuller / Academind

Updated Apr 2026; 16 sections / 263 lectures

<https://www.udemy.com/course/docker-kubernetes-the-practical-guide/>

Modern Event-Driven Spring Boot: Kafka & Spring Cloud Stream - Vinoth Selvaraj

Updated Jun 2026; Spring Boot 4 / Kafka 4; 20 sections / 190 lectures

<https://www.udemy.com/course/kafka-java/>

[NEW] Master Microservices with SpringBoot, Docker, Kubernetes - Eazy Bytes / Madan Reddy

Updated Aug 2026; 21 sections / 282 lectures

<https://www.udemy.com/course/master-microservices-with-spring-docker-kubernetes/>

GitHub Actions: The Complete Guide from Beginner to Expert - Lauro Fialho Muller

Updated Aug 2026; 20 sections / 115 lectures

<https://www.udemy.com/course/mastering-github-actions-beginner-to-expert/>

Jenkins: Jobs, Pipelines, CI/CD and DevOps for Beginners - Valentin Despa

Current 2026 catalog; 7 sections / 118 lectures

<https://www.udemy.com/course/jenkins-ci-cd-pipelines-devops-for-beginners/>

Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek

Updated Aug 2026; 33 sections / 397 lectures

<https://www.udemy.com/course/aws-certified-solutions-architect-associate-saa-c03/>

From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

<https://www.udemy.com/course/java-spring-ai/>

BFS Pass 1 - Breadth and baseline (Days 1-10)

Touch all tracks, establish mental maps, and identify RED/YELLOW/GREEN areas.

BFS Pass 2 - Internals and trade-offs (Days 11-20)

Return to every track at deeper failure, consistency, internals, and performance depth.

BFS Pass 3 - Production integration (Days 21-27)

Connect backend, data, frontend, cloud, CI/CD and AI into production systems.

Interview mode (Days 28-30)

No passive learning: timed coding, mock interviews, unknown system design, incident response and final scorecards.

DAY 1 - BFS Level 1 - Establish the map

Day goal: Touch every major track once and establish a baseline. Do not chase mastery today.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Arrays + HashMap

Assigned problems:

- Two Sum
- Group Anagrams
- Top K Frequent Elements

What to learn / focus on:

- HashMap one-pass patterns
- frequency maps
- heap vs bucket trade-off

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Modern Java - Collections + Functional Java

Course: Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

Course section / chapter to open:

- Course Specifics -> Exception Handling, IO, Collections, Generics, and Concurrency
- Functional-style Programming using Java 8 Features (Lambdas, Streams)
- Modern Java Features (Records / Pattern Matching / Switch Expressions)

What to learn / focus on:

- Collections choice by access pattern
- HashMap hashing/collision/load factor/treeification
- equals/hashCode contract
- generics and PECS
- lambdas and Streams: filter/map/flatMap/reduce/collect/groupingBy
- records as immutable data carriers

Skip / avoid in this block:

- Java installation, first program, primitive basics, loops and introductory OOP.

Hands-on work:

- Create a Transaction record; filter COMPLETED transactions; group by account; sum amount per account.
- Inspect PAT-FOS transaction state-machine `Map<Status,Set<Status>>` and explain why it is a graph adjacency map.

Interview questions you must answer:

- What happens inside HashMap.put()?
- ArrayList vs LinkedList?
- HashMap vs ConcurrentHashMap?
- map() vs flatMap()?
- When should Streams be avoided?

EXIT CRITERIA: Be able to choose ArrayList/HashMap/Set/TreeMap/PriorityQueue/ConcurrentHashMap from requirements and write a non-trivial Stream pipeline.

1:00 PM - 3:00 PM System Design - Framework + Payment System

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Design Interview Strategies -> Working Backwards
- Design Interview Strategies -> Defining Requirements
- Design Interview Strategies -> Design Strategies
- Designing Systems that Scale -> horizontal scaling / failover / caching

What to learn / focus on:

- Functional vs non-functional requirements
- rough scale/QPS
- API and data model
- high-level architecture
- deep dive/failure handling/trade-offs
- payment states, idempotency, provider timeout, webhook, reconciliation, outbox, ledger

Skip / avoid in this block:

- Do not start by naming Kafka/Redis before requirements justify them.

Hands-on work:

- Write payment APIs and state machine.
- Draw Client -> API Gateway -> Payment Service -> PostgreSQL -> Provider, then add components only when justified.
- Explain duplicate click, provider timeout, DB+Kafka dual-write, ledger.

Interview questions you must answer:

- Why 202/PENDING?
- Why idempotency key?
- What if provider charged but response was lost?
- What if DB commits but Kafka publish fails?

EXIT CRITERIA: Draw a first-pass Stripe/Temu payment architecture without looking at notes and explain four failure scenarios.

3:00 PM - 5:00 PM Data - SQL vs NoSQL + PostgreSQL fundamentals

Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- Simple - But Powerful - SQL Statements
- Database Design / Database Terminology / Creating Tables
- Index concepts later in course - use as preview

What to learn / focus on:

- relational modeling
- PK/FK/UNIQUE constraints
- normalization
- B-tree intuition
- when PostgreSQL vs MongoDB vs Redis vs Kafka

Skip / avoid in this block:

- Basic SELECT syntax drills you already know.

Hands-on work:

- Use PAT-FOS users/accounts/transactions schema. Decide indexes for from_account_id, status, idempotency_key.
- Explain why UNIQUE(idempotency_key) is a correctness constraint, not only a performance feature.

Interview questions you must answer:

- Why PostgreSQL for payments?
- When is MongoDB a better fit?
- Why is Kafka not the primary transaction database?

EXIT CRITERIA: Explain the storage role of PostgreSQL, MongoDB, Redis and Kafka in one minute each.

6:00 PM - 8:00 PM TypeScript + React - fast-track refresh

Course: Mastering TypeScript - 2026 Edition - Colt Steele

Updated Jun 2026; 19 sections / 151 lectures

Course section / chapter to open:

- Introduction -> The Case For TypeScript
- Introduction -> TypeScript Overview
- Type system / Union Types / Interfaces
- React course -> React Essentials - Deep Dive (selected summary material)

What to learn / focus on:

- type inference
- object/function types
- unions
- interfaces vs type aliases
- components/props/state/events/hooks refresher

Skip / avoid in this block:

- TypeScript installation/setup unless tooling is broken.
- Do not watch React 71h sequentially; cherry-pick Essentials.

Hands-on work:

- Type a PaymentStatus union and backend DTO response.
- Trace Spring DTO -> JSON -> Axios -> TypeScript -> React state -> component.

Interview questions you must answer:

- interface vs type?
- any vs unknown?
- props vs state?
- Why not mutate React state?

EXIT CRITERIA: Have one typed backend response flowing into a React component.

8:00 PM - 9:00 PM Docker - Level 1

Course: Docker & Kubernetes: The Practical Guide - Maximilian Schwarzmuller / Academind

Updated Apr 2026; 16 sections / 263 lectures

Course section / chapter to open:

- Docker Images & Containers: The Core Building Blocks -> Images & Containers: What and Why?
- Managing Images & Containers
- A Look Behind the Scenes: Inspecting Images

What to learn / focus on:

- image vs container vs VM
- Dockerfile layers
- build/run/ports

- multi-stage build purpose

Skip / avoid in this block:

- Beginner Docker installation material if Docker Desktop already works.

Hands-on work:

- Open PAT-FOS Spring Dockerfile and explain every line. Identify build stage vs runtime stage.

Interview questions you must answer:

- Why multi-stage?
- What creates a new image layer?
- What should never be baked into an image?

EXIT CRITERIA: Explain your Java Dockerfile end-to-end.

9:00 PM - 10:00 PM AI - Level 1 for Java

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Spring AI - Say Hello to AI in Spring Boot -> What is Spring AI Framework
- Hello World Spring AI app with OpenAI
- Generative AI / LLM fundamentals (token, context, embeddings preview)

What to learn / focus on:

- LLM
- token
- context window
- embedding
- ChatModel / ChatClient abstraction
- provider switching

Skip / avoid in this block:

- RAG and agents today - only preview terminology.

Hands-on work:

- Create/inspect a simple Spring AI REST endpoint and draw React -> Spring Boot -> Spring AI -> model provider.

Interview questions you must answer:

- What problem does Spring AI solve?
- Token vs embedding?
- Why use local Ollama vs hosted provider?

EXIT CRITERIA: Explain LLM/token/context/embedding and Spring AI in plain English.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 2 - BFS Level 1 - Concurrency, APIs and data flow

Day goal: Refresh concurrency and learn to reason about sliding windows, APIs, data access and AI retrieval.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Two Pointers + Sliding Window

Assigned problems:

- Valid Palindrome
- Two Sum II
- Longest Substring Without Repeating Characters

What to learn / focus on:

- left/right pointer invariants
- fixed vs variable window
- O(n) proof

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Java Concurrency - first pass

Course: Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

Course section / chapter to open:

- Course Specifics -> Concurrency (including Virtual Threads)
- Modern Java Features -> Virtual Threads

What to learn / focus on:

- thread lifecycle
- race condition
- atomicity vs visibility
- synchronized
- volatile
- AtomicInteger/CAS
- ExecutorService
- CompletableFuture
- virtual threads overview

Skip / avoid in this block:

- Do not memorize API methods without understanding the underlying problem.

Hands-on work:

- Write an unsafe concurrent counter, reproduce lost updates, then fix with AtomicInteger and synchronized.
- Create a small CompletableFuture fan-out example.

Interview questions you must answer:

- volatile vs synchronized?
- What is a race condition?
- Why can a thread pool improve throughput?
- When do virtual threads help?

EXIT CRITERIA: Be able to explain concurrency correctness with a concrete race and two fixes.

1:00 PM - 3:00 PM System Design - URL Shortener

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Design Interview Strategies -> Defining Requirements
- Mock Design Interviews -> Design a URL Shortening Service: Q&A / Try It Yourself / System Design / Debrief
- Designing Systems that Scale -> caching / sharding

What to learn / focus on:

- write-heavy vs read-heavy
- short-code generation
- collision handling
- redirect API
- database key/value model
- Redis caching
- TTL
- horizontal scaling

Skip / avoid in this block:

- Do not copy the course architecture before your own 45-minute attempt.

Hands-on work:

- Design requirements, APIs, capacity estimate, key-generation strategy, DB and cache.
- Compare random base62 IDs vs counter/hash.

Interview questions you must answer:

- How do you avoid collisions?
- How do you handle hot URLs?
- What if cache is down?

EXIT CRITERIA: Produce a complete URL shortener design and list three trade-offs.

3:00 PM - 5:00 PM PostgreSQL - joins and indexing introduction

Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- Simple SQL / relational data sections
- Joining Data from Different Resources
- Indexes material (preview and later revisit)

What to learn / focus on:

- INNER/LEFT joins
- join cardinality
- index selectivity
- single vs composite index
- read/write trade-off

Skip / avoid in this block:

- Beginner insert/update syntax if already fluent.

Hands-on work:

- Write PAT-FOS queries for account transaction history and owner lookup. Use EXPLAIN if local DB is running.

Interview questions you must answer:

- Why can an index make writes slower?
- When does a composite index help?
- What is selectivity?

EXIT CRITERIA: Choose indexes from actual query patterns rather than columns in isolation.

6:00 PM - 8:00 PM React Hooks + TypeScript functions

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- React Essentials - Deep Dive
- Dealing with side effects / built-in Hooks material
- TypeScript: function typing and unions

What to learn / focus on:

- useState
- useEffect
- cleanup
- dependency array mental model
- typed event handlers
- typed async function results

Skip / avoid in this block:

- CSS/styling modules today.

Hands-on work:

- Create a typed usePaymentStatus hook that polls an API and cleans up correctly.

Interview questions you must answer:

- When does useEffect run?
- What causes an infinite effect loop?
- Why cleanup?

EXIT CRITERIA: Build one custom hook with typed input/output and correct cleanup.

8:00 PM - 9:00 PM Docker - volumes and networking

Course: Docker & Kubernetes: The Practical Guide - Maximilian Schwarzmuller / Academind

Updated Apr 2026; 16 sections / 263 lectures

Course section / chapter to open:

- Managing Data & Working with Volumes
- Container networking material

What to learn / focus on:

- named volume vs bind mount
- container filesystem lifecycle
- bridge network / service name DNS
- environment variables

Skip / avoid in this block:

- Cloud deployment today.

Hands-on work:

- Inspect PAT-FOS docker-compose: name which data must persist and which services talk by hostname.

Interview questions you must answer:

- Volume vs bind mount?
- Why localhost is different inside a container?

EXIT CRITERIA: Explain how transaction-service reaches postgres/kafka/redis in Compose.

9:00 PM - 10:00 PM

AI - embeddings and semantic similarity

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Generative AI / LLM fundamentals -> Static Embeddings
- Positional Embeddings - How AI Understands Word Order
- RAG section -> Understanding the RAG flow (preview only)

What to learn / focus on:

- embedding vector intuition
- semantic vs lexical search
- cosine/dot-product intuition
- why embeddings enable retrieval

Skip / avoid in this block:

- Advanced RAG implementation today.

Hands-on work:

- Create 5 short payment-related sentences and reason which ones should be semantically nearest.

Interview questions you must answer:

- Embedding vs token?
- Why can vector search find synonyms?

EXIT CRITERIA: Explain how a user query becomes a vector and finds relevant documents.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 3 - BFS Level 1 - Spring internals, notification architecture and cache

Day goal: Understand Spring IoC at a deeper level while adding cache and event-driven concepts.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM **LeetCode / DSA**

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Stack + Queue

Assigned problems:

- Valid Parentheses
- Min Stack
- Daily Temperatures

What to learn / focus on:

- LIFO/FIFO
- monotonic stack intuition
- amortized complexity

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM **Spring Core - IoC, DI and auto-configuration**

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- Spring Boot 4 Quick Start -> Spring Boot Overview
- Spring Core -> What is Inversion of Control?
- Spring Core -> Defining Dependency Injection
- Spring Core Java configuration / component scanning material

What to learn / focus on:

- ApplicationContext
- bean definition
- component scan
- constructor injection
- bean lifecycle concept
- Spring Boot starters
- auto-configuration conditions

Skip / avoid in this block:

- First REST controller coding if too basic.

Hands-on work:

- Trace PAT-FOS TransactionService dependencies; draw which beans Spring creates and injects.
- Replace one field-injection example (if any) with constructor injection.

Interview questions you must answer:

- IoC vs DI?
- How does Spring know what to inject?
- Why constructor injection?
- What is auto-configuration?

EXIT CRITERIA: Explain the path from @SpringBootApplication to a constructed service bean.

1:00 PM - 3:00 PM System Design - Notification Service

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Working with Big Data -> Message Queues
- Designing Systems that Scale -> caching/failover
- Design Interview Strategies

What to learn / focus on:

- email/SMS/push channels
- producer API
- Kafka/SQS queue
- worker pools
- preferences
- rate limits
- retry/backoff
- DLQ
- idempotency

Skip / avoid in this block:

- Do not make notification sending synchronous with the business request.

Hands-on work:

- Design notification event schema and worker flow. Add provider failure and retry strategy.

Interview questions you must answer:

- Why queue?
- How prevent duplicate notification?
- How handle provider throttling?

EXIT CRITERIA: Explain async notification flow and failure handling.

3:00 PM - 5:00 PM Redis - cache-aside first pass

Course: Redis: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; Redis 7; 22 sections / 187 lectures

Course section / chapter to open:

- Caching implementation material -> Redis Design Methodology
- Key Naming Methodology
- Adding Page Caching
- Redis core data structures overview

What to learn / focus on:

- GET/SET
- TTL
- cache-aside
- key naming
- cache invalidation

- strings/hashtes/sorted sets use cases

Skip / avoid in this block:

- Local Redis installation if already running in Docker.

Hands-on work:

- Inspect PAT-FOS balances::<accountId>. Trace miss -> DB -> cache and write -> eviction.
- Implement a simple rate-limit counter with INCR + EXPIRE.

Interview questions you must answer:

- What if Redis is down?
- Cache-aside vs write-through?
- Why is cache invalidation hard?

EXIT CRITERIA: Be able to design Redis as an optimization while PostgreSQL stays source of truth.

6:00 PM - 8:00 PM React forms + TypeScript narrowing

Course: Mastering TypeScript - 2026 Edition - Colt Steele

Updated Jun 2026; 19 sections / 151 lectures

Course section / chapter to open:

- Type Narrowing
- Interfaces / unions sections
- React -> handling forms and user input (selected)
- React Form Actions overview if available

What to learn / focus on:

- discriminated unions
- narrowing with in/typeof
- controlled inputs
- form validation
- typed form model

Skip / avoid in this block:

- Advanced Redux today.

Hands-on work:

- Create a TransferForm type with success/error/pending states as a discriminated union.

Interview questions you must answer:

- What is narrowing?
- Controlled vs uncontrolled input?
- How do union types prevent impossible UI states?

EXIT CRITERIA: Build a typed form with at least three validation rules.

8:00 PM - 9:00 PM CI/CD foundations

Course: Jenkins: Jobs, Pipelines, CI/CD and DevOps for Beginners - Valentin Despa

Current 2026 catalog; 7 sections / 118 lectures

Course section / chapter to open:

- Jenkins, CI/CD and DevOps fundamentals
- Implementing Continuous Integration (CI) with Jenkins -> Introduction to CI
- How to structure a pipeline?

What to learn / focus on:

- CI vs CD
- pipeline-as-code
- stages
- fail-fast
- artifact

- test report

Skip / avoid in this block:

- Jenkins installation details unless needed.

Hands-on work:

- Draw PAT-FOS pipeline: checkout -> Java/Python/frontend tests -> integration -> image build.

Interview questions you must answer:

- Why CI?
- Why separate stages?
- What should stop deployment?

EXIT CRITERIA: Explain a CI pipeline from commit to verified artifact.

9:00 PM - 10:00 PM AI - vector DB + RAG map

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- The Art of Talking to Documents - RAG Unleashed -> From Prompt Stuffing to RAG
- Understanding the RAG flow

What to learn / focus on:

- ingestion
- chunk
- embed
- vector store
- retrieve
- augment prompt
- generate

Skip / avoid in this block:

- Qdrant setup implementation until Day 9/27.

Hands-on work:

- Draw a RAG flow over PAT-FOS docs and define metadata fields (source, section, timestamp).

Interview questions you must answer:

- Why RAG instead of putting all docs in prompt?
- What can retrieval get wrong?

EXIT CRITERIA: Explain six RAG stages without notes.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 4 - BFS Level 1 - Persistence, rate limiting and AWS basics

Day goal: Connect ORM/transactions with persistence choices and cloud primitives.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Linked List

Assigned problems:

- Reverse Linked List
- Merge Two Sorted Lists
- Linked List Cycle

What to learn / focus on:

- dummy nodes
- pointer rewiring
- Floyd cycle detection

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM JPA / Hibernate - first pass

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- Spring Boot REST CRUD / Hibernate-JPA sections
- JPA / Hibernate Advanced Mappings -> Overview Part 1 / Part 2

What to learn / focus on:

- entity states
- persistence context
- dirty checking
- lazy/eager loading
- one-to-many/many-to-one
- N+1 query problem
- DTO projection concept

Skip / avoid in this block:

- Introductory MySQL setup if PAT-FOS Postgres is available.

Hands-on work:

- Inspect Account -> User and Transaction -> Accounts mappings. Identify lazy relations and possible N+1 query.

Interview questions you must answer:

- What is persistence context?

- What causes LazyInitializationException?
- Why is EAGER not a universal fix?

EXIT CRITERIA: Explain one request from repository query through Hibernate to SQL and back.

1:00 PM - 3:00 PM System Design - Rate Limiter

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Designing Systems that Scale -> Caching / Eviction Strategies
- Algorithms and Data Structures -> Hash Tables
- Design Interview Strategies

What to learn / focus on:

- token bucket
- leaky bucket
- fixed/sliding window
- Redis atomic operations
- distributed limiters
- per-user vs per-IP

Skip / avoid in this block:

- Do not over-focus on API Gateway products before algorithm semantics.

Hands-on work:

- Design 100 requests/min/user. Define Redis keys and atomic update. Compare sliding-window log vs token bucket.

Interview questions you must answer:

- Race conditions across servers?
- What if Redis fails?
- How enforce global limit multi-region?

EXIT CRITERIA: Defend one algorithm and state its memory/accuracy trade-off.

3:00 PM - 5:00 PM PostgreSQL - ACID and transactions first pass

Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- Transaction/concurrency material later in course
- Database Design sections
- use PAT-FOS DB for hands-on

What to learn / focus on:

- ACID
- commit/rollback
- isolation overview
- lost update
- dirty/non-repeatable/phantom read
- row locks

Skip / avoid in this block:

- Memorizing isolation table without reproducing an anomaly.

Hands-on work:

- Run two concurrent transfer transactions or simulate sessions; observe SELECT FOR UPDATE behavior.

Interview questions you must answer:

- Why is atomicity critical for debit+credit?
- Isolation vs consistency?

EXIT CRITERIA: Explain how DB transaction protects a financial transfer.

6:00 PM - 8:00 PM React Router + API integration

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- Routing with React Router
- Sending HTTP requests & handling transitional states + responses

What to learn / focus on:

- route configuration
- protected route concept
- loader/data fetching overview
- Axios/fetch error/loading states

Skip / avoid in this block:

- Next.js today.

Hands-on work:

- Create routes for dashboard/history/admin and typed API client calls.

Interview questions you must answer:

- SPA route vs server route?
- How do you represent loading/error/success?

EXIT CRITERIA: Navigate PAT-FOS pages and trace API call lifecycle.

8:00 PM - 9:00 PM AWS - Regions, IAM, EC2, ALB overview

Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- Getting started with AWS -> AWS Cloud Overview - Regions & AZ
- EC2 fundamentals (selected)
- ELB / Auto Scaling sections (selected)

What to learn / focus on:

- Region/AZ
- IAM role vs user
- EC2 instance
- security group
- ALB
- Auto Scaling Group

Skip / avoid in this block:

- Account creation/setup if already configured.

Hands-on work:

- Map local transaction-service to EC2/ECS conceptually; draw ALB -> multiple stateless instances.

Interview questions you must answer:

- Region vs AZ?
- Why IAM role on EC2 instead of access keys?

EXIT CRITERIA: Explain how horizontal scaling works on AWS.

9:00 PM - 10:00 PM AI - RAG architecture first pass

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- The Art of Talking to Documents - RAG Unleashed -> Understanding the RAG flow
- Breaking Down Big Documents - The Art of Chunking for LLMs

What to learn / focus on:

- chunk size/overlap
- metadata
- top-k
- similarity threshold
- grounding

Skip / avoid in this block:

- Advanced reranking today.

Hands-on work:

- Choose chunk boundaries for service-tour.md and Q&A-interview.md; note what metadata to preserve.

Interview questions you must answer:

- What happens if chunks are too small/large?
- Why metadata filters?

EXIT CRITERIA: Create a concrete ingestion plan for PAT-FOS documents.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 5 - BFS Level 1 - REST, security, NoSQL and AI structured output

Day goal: Cover application boundaries: secure REST, flexible data and typed AI responses.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Binary Search

Assigned problems:

- Binary Search
- Search in Rotated Sorted Array
- Find Minimum in Rotated Sorted Array

What to learn / focus on:

- search-space invariant
- boundary conditions
- $O(\log n)$ reasoning

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Spring REST + Security - first pass

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- Spring Boot REST API CRUD / OpenAPI and Swagger
- Spring Boot REST API Security -> JDBC Authentication / BCrypt / custom tables

What to learn / focus on:

- REST resource design
- status codes
- validation
- ProblemDetail
- SecurityFilterChain
- authentication vs authorization
- BCrypt
- RBAC

Skip / avoid in this block:

- Spring MVC Thymeleaf security.

Hands-on work:

- Trace PAT-FOS JWT filter -> SecurityContext -> @PreAuthorize. Add/test one 403 case.

Interview questions you must answer:

- 401 vs 403?
- Why BCrypt?
- Why stateless JWT?
- Where should validation live?

EXIT CRITERIA: Explain auth flow and error mapping end-to-end.

1:00 PM - 3:00 PM System Design - Order Management System

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Design Interview Strategies
- Working with Big Data -> Message Queues
- Designing Systems that Scale -> failover/caching

What to learn / focus on:

- order lifecycle
- inventory reservation
- payment coordination
- idempotency
- saga
- events
- eventual consistency

Skip / avoid in this block:

- Pretending a distributed transaction spans all services.

Hands-on work:

- Design order states and events. Add compensation for inventory reserved but payment failed.

Interview questions you must answer:

- Saga choreography vs orchestration?
- How prevent duplicate order creation?

EXIT CRITERIA: Explain order/payment/inventory failure path.

3:00 PM - 5:00 PM MongoDB - document model first pass

Course: MongoDB - The Complete Developer's Guide - Maximilian Schwarzmuller / Academind

Updated Jan 2026; 19 sections / 266 lectures

Course section / chapter to open:

- MongoDB basics / CRUD modules
- Data modeling material
- Working with Indexes (preview)

What to learn / focus on:

- document vs row
- embedding vs referencing
- schema flexibility
- ObjectId
- basic index choice

Skip / avoid in this block:

- Installation if Mongo Express/local Mongo already works.

Hands-on work:

- Inspect fraud_assessments and transaction_events. Decide what should be embedded vs referenced.

Interview questions you must answer:

- When embed?

- When reference?
- Why not store primary ledger in MongoDB?

EXIT CRITERIA: Design a fraud assessment document with useful indexes.

6:00 PM - 8:00 PM React authentication + TypeScript state model

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmüller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- Implementing Authentication
- React Router protected routing material
- TypeScript -> union types / interfaces

What to learn / focus on:

- token storage trade-offs
- auth context
- role-based UI
- typed auth state
- logout/expiry

Skip / avoid in this block:

- Next.js auth.

Hands-on work:

- Model AuthState as discriminated union; guard admin route and handle expired token.

Interview questions you must answer:

- localStorage risks?
- 401 handling?
- front-end authorization vs backend authorization?

EXIT CRITERIA: Secure UI navigation while recognizing backend remains authoritative.

8:00 PM - 9:00 PM AWS - IAM + VPC mental model

Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- IAM sections (selected)
- VPC fundamentals (selected)
- EC2 security groups recap

What to learn / focus on:

- least privilege
- role/policy
- public/private subnet
- route table
- security group vs NACL

Skip / avoid in this block:

- Deep NAT/VPN design today.

Hands-on work:

- Draw public ALB and private app/database tiers for PAT-FOS.

Interview questions you must answer:

- Security group stateful?
- Why DB private?

EXIT CRITERIA: Explain network isolation for a production backend.

9:00 PM - 10:00 PM AI - prompts + structured Java output**Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy**

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Spring AI sections on prompt templates / response conversion
- Spring AI - Say Hello to AI in Spring Boot (ChatClient usage)

What to learn / focus on:

- system/user prompts
- prompt template variables
- structured output to Java record/list/map
- validation of model output

Skip / avoid in this block:

- Agents/MCP.

Hands-on work:

- Define Java record FraudExplanation and prompt model to fill fields reason, confidence, nextAction.

Interview questions you must answer:

- Why structured output?
- What if model returns invalid shape?

EXIT CRITERIA: Return one typed Java object from an AI call.**End-of-day checkpoint**

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 6 - BFS Level 1 - Transactions, task system and Kubernetes basics

Day goal: Connect transaction boundaries with task-system architecture and container orchestration.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Trees

Assigned problems:

- Maximum Depth of Binary Tree
- Binary Tree Level Order Traversal
- Validate Binary Search Tree

What to learn / focus on:

- DFS recursion
- BFS queue
- BST range invariant

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Spring @Transactional - first pass

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- Spring/JPA transaction material (course + hands-on PAT-FOS)
- JPA / Hibernate mappings as context

What to learn / focus on:

- transaction proxy concept
- commit/rollback
- RuntimeException rollback default
- propagation overview
- pessimistic vs optimistic lock

Skip / avoid in this block:

- Long propagation enum memorization.

Hands-on work:

- Add a failure between debit and credit in a test and verify rollback.

Interview questions you must answer:

- Does @Transactional work on private/self-invoked method? (preview for Day12)
- REQUIRED vs REQUIRES_NEW?

EXIT CRITERIA: Demonstrate atomic rollback in a test.

1:00 PM - 3:00 PM System Design - Task Management System

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Design Interview Strategies
- Designing Systems that Scale -> caching / sharding / resiliency

What to learn / focus on:

- projects/tasks/subtasks/comments
- API
- schema
- search
- notifications
- attachments
- real-time updates
- rate limiting

Skip / avoid in this block:

- Reading your existing Q&A answer before first attempt.

Hands-on work:

- Design from scratch; afterward compare against PAT-FOS Q&A Task Management design.

Interview questions you must answer:

- Cursor vs offset pagination?
- Why Elasticsearch for search?
- How store attachments?

EXIT CRITERIA: Produce complete task-system architecture and comparison notes.

3:00 PM - 5:00 PM PostgreSQL - indexing deeper

Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- Indexes sections (course deep dive)
- EXPLAIN / query-plan material
- PAT-FOS index experiments

What to learn / focus on:

- B-tree
- compound index leftmost prefix
- covering/partial index concept
- EXPLAIN
- selectivity

Skip / avoid in this block:

- Creating indexes without measuring query.

Hands-on work:

- EXPLAIN transaction history queries before/after index.

Interview questions you must answer:

- Index (project_id,status) vs (status,project_id)?
- When can index be ignored?

EXIT CRITERIA: Read a query plan at a basic level and justify index order.

6:00 PM - 8:00 PM React Context + shared state**Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind**

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- React Context API / managing cross-component state
- React Essentials state management recap
- TypeScript interfaces for context

What to learn / focus on:

- provider/consumer
- context re-render cost
- local vs shared state
- auth/account context

Skip / avoid in this block:

- Redux until Day7/19.

Hands-on work:

- Create typed AuthContext or AccountContext and remove one prop-drilling chain.

Interview questions you must answer:

- When is Context enough?
- Why can large Context cause re-renders?

EXIT CRITERIA: Use Context intentionally, not as default global state.**8:00 PM - 9:00 PM Kubernetes - Pod/Deployment/Service****Course: Docker & Kubernetes: The Practical Guide - Maximilian Schwarzmuller / Academind**

Updated Apr 2026; 16 sections / 263 lectures

Course section / chapter to open:

- Kubernetes core concepts & architecture
- Creating Kubernetes resources, deployments, services

What to learn / focus on:

- Pod
- Deployment
- ReplicaSet
- Service
- declarative desired state

Skip / avoid in this block:

- Volumes/Ingress/HPA until later.

Hands-on work:

- Write minimal Deployment + ClusterIP Service YAML for transaction-service.

Interview questions you must answer:

- Pod vs container?
- Deployment vs Pod?
- Why Service?

EXIT CRITERIA: Explain request routing to replicated pods.**9:00 PM - 10:00 PM Spring AI - first real endpoint****Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy**

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Spring AI - Say Hello to AI in Spring Boot -> Hello World OpenAI/Ollama/ Docker model runner

What to learn / focus on:

- ChatClient fluent API
- provider configuration
- API key as secret
- local vs hosted model

Skip / avoid in this block:

- RAG today.

Hands-on work:

- Build /api/ai/explain?question= endpoint locally or pseudocode if credentials unavailable.

Interview questions you must answer:

- How switch providers?
- What data should never go to public LLM?

EXIT CRITERIA: Have a working or fully specified Spring AI endpoint.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 7 - BFS Level 1 - Graphs, Kafka and state management

Day goal: Strengthen graph reasoning and add event-streaming fundamentals.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Graphs BFS / DFS

Assigned problems:

- Number of Islands
- Clone Graph
- Course Schedule

What to learn / focus on:

- visited sets
- cycle detection
- topological ordering

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Java Collections Internals - deeper first pass

Course: Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

Course section / chapter to open:

- Course Specifics -> Exception Handling, IO, Collections, Generics, and Concurrency
- Functional-style Programming using Java 8 Features (Lambdas, Streams)
- Modern Java Features (Records / Pattern Matching / Switch Expressions)

What to learn / focus on:

- HashMap bucket mechanics
- ConcurrentHashMap concurrency idea
- PriorityQueue heap
- ArrayDeque
- immutable collections
- record equality semantics

Skip / avoid in this block:

- Basic List/Set syntax.

Hands-on work:

- Implement small LRU-like structure with LinkedHashMap or explain design; compare PriorityQueue use in Top K.

Interview questions you must answer:

- Why CHM not synchronized HashMap?

- Heap operations complexity?

EXIT CRITERIA: Connect collection internals to LeetCode and production choices.

1:00 PM - 3:00 PM System Design - Search / Autocomplete

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Algorithms and Data Structures -> Information Retrieval
- Designing Systems that Scale -> caching/sharding

What to learn / focus on:

- prefix search
- Trie
- inverted index
- ranking
- precomputation
- Redis cache
- event ingestion

Skip / avoid in this block:

- Trying to make PostgreSQL LIKE query the only large-scale solution.

Hands-on work:

- Design autocomplete API and data pipeline for trending suggestions.

Interview questions you must answer:

- Trie vs Elasticsearch?
- How update suggestions in near-real-time?

EXIT CRITERIA: Explain read path and update path separately.

3:00 PM - 5:00 PM Kafka - core fundamentals

Course: Modern Event-Driven Spring Boot: Kafka & Spring Cloud Stream - Vinoth Selvaraj

Updated Jun 2026; Spring Boot 4 / Kafka 4; 20 sections / 190 lectures

Course section / chapter to open:

- Introduction To Event Driven Architecture
- Kafka Core Fundamentals -> Kafka Core Concepts
- consumer groups / ordering / partition rebalancing / offset tracking

What to learn / focus on:

- topic
- partition
- offset
- broker/controller
- producer key
- consumer group
- ordering
- lag
- rebalance

Skip / avoid in this block:

- Spring Cloud Stream abstraction before understanding core Kafka.

Hands-on work:

- Use Kafka UI to inspect PAT-FOS topics, partitions, offsets and consumer groups.

Interview questions you must answer:

- What ordering does Kafka guarantee?

- Why partitions limit parallel consumers?
- What is lag?

EXIT CRITERIA: Draw Kafka internals for one topic with 3 partitions and 2 consumers.

6:00 PM - 8:00 PM Redux Toolkit - first pass

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- Redux & Redux Toolkit
- React Context recap

What to learn / focus on:

- store
- slice
- reducer
- action
- selector
- async thunk concept
- when Redux is justified

Skip / avoid in this block:

- Legacy Redux boilerplate.

Hands-on work:

- Model transaction history/filter state in a slice or outline it; compare to Context.

Interview questions you must answer:

- Redux vs Context?
- Why immutable updates?

EXIT CRITERIA: Defend state-management choice rather than using Redux by habit.

8:00 PM - 9:00 PM Kubernetes - ConfigMap, Secret, Ingress

Course: Docker & Kubernetes: The Practical Guide - Maximilian Schwarzmuller / Academind

Updated Apr 2026; 16 sections / 263 lectures

Course section / chapter to open:

- Kubernetes networking / DNS service discovery
- Kubernetes configuration / secrets / ingress material

What to learn / focus on:

- ConfigMap
- Secret
- Ingress
- Service DNS
- config separation

Skip / avoid in this block:

- Helm until later.

Hands-on work:

- Create ConfigMap for app config and Secret placeholder for DB password.

Interview questions you must answer:

- Secret vs ConfigMap?
- Why not put secrets in image?

EXIT CRITERIA: Explain external request -> Ingress -> Service -> Pod.

9:00 PM - 10:00 PM AI - embeddings with Spring AI**Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy**

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- The Art of Talking to Documents - RAG Unleashed -> vector store setup concepts
- Static Embeddings / RAG flow

What to learn / focus on:

- EmbeddingModel
- vector store abstraction
- document + metadata
- similarity search

Skip / avoid in this block:

- Full ingestion pipeline until Day9.

Hands-on work:

- Sketch Java code that embeds Transaction FAQ documents and stores them.

Interview questions you must answer:

- What is topK?
- Similarity threshold?

EXIT CRITERIA: Know the Spring AI components needed for retrieval.**End-of-day checkpoint**

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 8 - BFS Level 1 - JVM performance, Kafka ordering and AWS containers

Day goal: Connect runtime performance to event-streaming and cloud deployment.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Heap / Priority Queue

Assigned problems:

- Kth Largest Element in an Array
- K Closest Points to Origin
- Task Scheduler

What to learn / focus on:

- min vs max heap
- top-k pattern
- scheduling trade-offs

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM JVM + application performance

Course: Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

Course section / chapter to open:

- Course Specifics -> JVM Internals
- Java: A High-level Overview -> Java Virtual Machine (JVM)
- Java SE Release Cycles & Release History

What to learn / focus on:

- stack vs heap
- class loading
- JIT intuition
- GC goals
- allocation pressure
- thread pools vs DB pool
- latency vs throughput

Skip / avoid in this block:

- GC collector flag memorization.

Hands-on work:

- Inspect PAT-FOS JVM memory/threads via actuator/JVM tools if available; identify external bottlenecks before tuning JVM.

Interview questions you must answer:

- Memory leak in managed language?
- CPU vs I/O bound?
- Why bigger thread pool can hurt?

EXIT CRITERIA: Explain a systematic Spring Boot performance investigation.

1:00 PM - 3:00 PM System Design - News Feed

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Designing Systems that Scale -> caching/sharding/failover
- Working with Big Data -> Message Queues

What to learn / focus on:

- fanout-on-write vs fanout-on-read
- celebrity problem
- feed cache
- ranking
- event queue
- pagination

Skip / avoid in this block:

- One strategy for every user size.

Hands-on work:

- Design hybrid feed generation and explain hot-celebrity handling.

Interview questions you must answer:

- Why Redis sorted set?
- How paginate feed?
- How handle deletes?

EXIT CRITERIA: Defend hybrid fanout.

3:00 PM - 5:00 PM Kafka - keys, partitioning and ordering

Course: Modern Event-Driven Spring Boot: Kafka & Spring Cloud Stream - Vinoth Selvaraj

Updated Jun 2026; Spring Boot 4 / Kafka 4; 20 sections / 190 lectures

Course section / chapter to open:

- Kafka Core Fundamentals -> Why Message Ordering Matters
- Consumer Group Scaling Scenarios
- Partition Rebalancing
- Offset Tracking Deep Dive

What to learn / focus on:

- key -> partition
- per-key ordering
- changing partition count risk
- rebalance impact
- offset commit concept

Skip / avoid in this block:

- Transactions/exactly-once until Day13.

Hands-on work:

- For PAT-FOS, compare key=transactionId vs key=accountId. State ordering/parallelism consequences.

Interview questions you must answer:

- Can Kafka guarantee global order?
- What happens when partitions increase?

EXIT CRITERIA: Choose correct partition key for three concrete use cases.

6:00 PM - 8:00 PM React HTTP + Axios architecture

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- Sending Http requests & handling transitional states + responses
- React Essentials side effects recap
- Combining React with TypeScript

What to learn / focus on:

- central API client
- interceptors
- typed response
- error normalization
- cancellation/cleanup
- retry caution

Skip / avoid in this block:

- Visual styling.

Hands-on work:

- Refactor PAT-FOS transactionApi.ts to typed functions and centralized auth/error handling.

Interview questions you must answer:

- Where should retry live?
- What should happen on 401?

EXIT CRITERIA: Have a clean typed API boundary.

8:00 PM - 9:00 PM AWS - ECS/Fargate/Lambda first pass

Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- AWS Containers Services -> ECS / ECR / EKS / App Runner
- Serverless -> Lambda overview

What to learn / focus on:

- ECR image registry
- ECS cluster/service/task definition
- Fargate
- Lambda fit/non-fit

Skip / avoid in this block:

- EKS detail until Day22/25.

Hands-on work:

- Map Docker Compose transaction-service to ECS task + service + ALB.

Interview questions you must answer:

- ECS vs EKS?
- Fargate vs EC2?
- When Lambda not a fit?

EXIT CRITERIA: Explain deployment of Spring Boot container to ECS Fargate.

9:00 PM - 10:00 PM AI - embeddings implementation plan

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- RAG Unleashed -> Qdrant Vector Store setup / random-data RAG demos

What to learn / focus on:

- Qdrant or supported vector store
- document ingestion
- similarity request
- topK
- threshold

Skip / avoid in this block:

- Advanced RAG.

Hands-on work:

- Define data model for PAT-FOS doc chunks and create a minimal vector-store config.

Interview questions you must answer:

- Vector DB vs relational DB?
- Can Postgres pgvector be alternative?

EXIT CRITERIA: Ready to build RAG on Day9.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 9 - BFS Level 1 - Resilience, locking and first RAG

Day goal: Handle failures deliberately rather than only happy-path architecture.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Intervals

Assigned problems:

- Merge Intervals
- Insert Interval
- Non-overlapping Intervals

What to learn / focus on:

- sort then sweep
- overlap invariant
- greedy interval choice

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Spring microservice resilience - first pass

Course: [NEW] Master Microservices with SpringBoot,Docker,Kubernetes - Eazy Bytes / Madan Reddy

Updated Aug 2026; 21 sections / 282 lectures

Course section / chapter to open:

- Building resilient microservices using RESILIENCE4J framework -> circuit breaker / retry / rate limiter / bulkhead
- Aspect order of Resiliency patterns

What to learn / focus on:

- timeout
- retry with backoff/jitter
- circuit breaker states
- bulkhead
- rate limiter
- fallback

Skip / avoid in this block:

- Retrying non-idempotent payment blindly.

Hands-on work:

- Wrap a mock provider call with timeout + circuit breaker; define which errors are retryable.

Interview questions you must answer:

- Retry vs circuit breaker?
- Why jitter?

- Bulkhead purpose?

EXIT CRITERIA: Choose resilience policy by failure type.

1:00 PM - 3:00 PM System Design - Ticket / Reservation System

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Mock Design Interviews -> Restaurant Reservation System Q&A / Try It Yourself / System Design / Debrief
- Designing Systems that Scale -> ACID / CAP

What to learn / focus on:

- inventory/seat holds
- TTL
- overselling
- locking
- payment timeout
- idempotency

Skip / avoid in this block:

- Eventual consistency for final seat ownership without reservation semantics.

Hands-on work:

- Design hold -> payment -> confirm/expire state machine.

Interview questions you must answer:

- Optimistic vs pessimistic lock?
- How expire holds?

EXIT CRITERIA: Explain how you prevent double booking.

3:00 PM - 5:00 PM Database locking - optimistic vs pessimistic

Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- Transaction/concurrency material
- PAT-FOS SELECT FOR UPDATE implementation
- Mongo/Redis comparison only as needed

What to learn / focus on:

- pessimistic row lock
- optimistic version column
- deadlock ordering
- lock timeout
- high-contention trade-off

Skip / avoid in this block:

- Assuming locks solve distributed provider uncertainty.

Hands-on work:

- Run two simultaneous transfers from same account; document blocking behavior. Design optimistic alternative with retry.

Interview questions you must answer:

- When optimistic wins?
- How deadlocks arise?

EXIT CRITERIA: Defend locking strategy for payments vs profile edits.

6:00 PM - 8:00 PM React performance first pass**Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind**

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- React behind-the-scenes / DevTools material
- built-in Hooks -> useMemo/useCallback where appropriate

What to learn / focus on:

- render triggers
- memoization cost
- React.memo
- useMemo/useCallback
- list keys

Skip / avoid in this block:

- Premature memoization everywhere.

Hands-on work:

- Use React DevTools on PAT-FOS and find one avoidable re-render; optimize only if measurable.

Interview questions you must answer:

- useMemo vs useCallback?
- Why index key can be bad?

EXIT CRITERIA: Explain performance from render cause, not folklore.**8:00 PM - 9:00 PM GitHub Actions - workflow first pass****Course: GitHub Actions: The Complete Guide from Beginner to Expert - Lauro Fialho Muller**

Updated Aug 2026; 20 sections / 115 lectures

Course section / chapter to open:

- Introduction
- workflow building blocks / triggers / jobs / steps
- best practices for CI/CD pipelines

What to learn / focus on:

- workflow YAML
- on triggers
- job dependency
- runner
- cache/artifact
- secrets

Skip / avoid in this block:

- Custom actions/reusable workflow until Day23.

Hands-on work:

- Open PAT-FOS ci.yml and explain needs graph.

Interview questions you must answer:

- Job vs step?
- How run jobs in parallel?

EXIT CRITERIA: Explain CI workflow from push to Docker image.**9:00 PM - 10:00 PM AI - build first RAG****Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy**

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- RAG Unleashed -> Demo of RAG using random text data
- Let's talk to documents using RAG
- Breaking Down Big Documents

What to learn / focus on:

- ingest PAT-FOS docs
- chunk
- embed
- retrieve
- system prompt with context
- unknown-answer behavior

Skip / avoid in this block:

- Agents.

Hands-on work:

- Implement or outline /ai/ask-docs using 1-2 project markdown files. Test 5 questions.

Interview questions you must answer:

- How reduce hallucination?
- How measure retrieval quality?

EXIT CRITERIA: Produce a working first RAG or testable implementation plan with sample results.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 10 - BFS Pass 1 checkpoint

Day goal: Stop adding breadth for one day and prove what you retained from the first nine days.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Mixed BFS Pass-1 Mock

Assigned problems:

- 3 random medium problems from Days 1-9

What to learn / focus on:

- problem recognition
- communication
- testing and complexity

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Java + Spring Mock - breadth checkpoint

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course video unless a gap is exposed

What to learn / focus on:

- collections
- concurrency
- Spring IoC
- JPA
- REST/security
- transactions
- JVM performance

Skip / avoid in this block:

- Passive review.

Hands-on work:

- 60-min Java oral mock + 60-min Spring mock. For each miss, write exact course section to revisit in Pass 2.

Interview questions you must answer:

- Why equals/hashCode?
- volatile vs synchronized?
- IoC?
- N+1?
- 401 vs 403?

- transaction proxy?

EXIT CRITERIA: Create a ranked weakness list: RED / YELLOW / GREEN.

1:00 PM - 3:00 PM System Design - Payment System redo

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No reference for first 60 minutes; then compare with Mastering System Design sections

What to learn / focus on:

- requirements
- API
- data model
- provider
- idempotency
- webhook
- reconciliation
- outbox
- ledger
- scaling

Skip / avoid in this block:

- Looking at Day1 design before first attempt.

Hands-on work:

- 45-min mock + 15-min challenge; then compare with your Day1 notes and write 5 improvements.

Interview questions you must answer:

- What if provider times out?
- What if webhook duplicated?
- What if outbox publisher crashes?

EXIT CRITERIA: Produce Payment System v2 diagram and gap list.

3:00 PM - 5:00 PM Data checkpoint - SQL + Redis + Kafka + Mongo

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Use relevant course sections only to patch misses

What to learn / focus on:

- DB choice
- index choice
- locking
- cache invalidation
- Kafka ordering/lag
- Mongo modeling

Skip / avoid in this block:

- Rewatching whole sections.

Hands-on work:

- 20-min oral drill per technology; inspect live PAT-FOS stores with Adminer/Kafka UI/Mongo Express/redis-cli.

Interview questions you must answer:

- Source of truth?
- Failure behavior?
- Consistency model?

EXIT CRITERIA: One-page data architecture cheat sheet.

6:00 PM - 8:00 PM Frontend checkpoint - typed page from scratch**Course:** Active practice / mock - no required video before the first attempt**Course section / chapter to open:**

- React Essentials + TypeScript only if blocked

What to learn / focus on:

- typed API
- hooks
- routing
- auth
- forms
- performance basics

Skip / avoid in this block:

- Copy existing component.

Hands-on work:

- Build a small TransactionHistory page from blank file using typed mock/API data.

Interview questions you must answer:

- Where state lives?
- How handle loading/error?

EXIT CRITERIA: A working page without copying PAT-FOS implementation.**8:00 PM - 9:00 PM DevOps checkpoint - local to CI to cloud****Course:** Active practice / mock - no required video before the first attempt**Course section / chapter to open:**

- Docker / GitHub Actions / AWS notes

What to learn / focus on:

- Docker image
- Compose
- K8s basics
- CI stages
- AWS mapping

Skip / avoid in this block:

- Course video.

Hands-on work:

- Draw Git push -> CI -> image -> ECS -> RDS/Redis/MSK.

Interview questions you must answer:

- Where secrets?
- Rollback?

EXIT CRITERIA: Explain deployment flow in <5 minutes.**9:00 PM - 10:00 PM AI checkpoint - RAG explanation****Course:** Active practice / mock - no required video before the first attempt**Course section / chapter to open:**

- Spring AI notes only

What to learn / focus on:

- LLM
- token
- embedding

- RAG
- vector store
- Spring AI

Skip / avoid in this block:

- New AI topic.

Hands-on work:

- Explain to an interviewer; test RAG with 5 questions if implemented.

Interview questions you must answer:

- What can fail in RAG?

EXIT CRITERIA: Pass/fail self-score and prioritize Pass 2.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 11 - BFS Level 2 - Concurrency internals, MVCC and retrieval quality

Day goal: Revisit familiar concepts at internals/trade-off depth.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Graph BFS/DFS - deeper

Assigned problems:

- Pacific Atlantic Water Flow
- Rotting Oranges
- Surrounded Regions

What to learn / focus on:

- multi-source BFS
- reverse graph thinking
- grid complexity

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Java Concurrency - deeper

Course: Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

Course section / chapter to open:

- Course Specifics -> Concurrency (including Virtual Threads)
- Modern Java Features -> Virtual Threads

What to learn / focus on:

- Java Memory Model intuition
- happens-before
- visibility
- CAS
- atomic classes
- lock contention
- deadlock
- ConcurrentHashMap internals concept

Skip / avoid in this block:

- Intro thread creation.

Hands-on work:

- Build deadlock example then fix by lock ordering; benchmark coarse synchronized vs atomic counter qualitatively.

Interview questions you must answer:

- What guarantees visibility?
- What is ABA/CAS limitation conceptually?
- How avoid deadlock?

EXIT CRITERIA: Explain atomicity/visibility/ordering separately.

1:00 PM - 3:00 PM System Design - Payment failure deep dive

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Designing Systems that Scale -> Resiliency
- ACID compliance and CAP theorem
- Design Interview Strategies

What to learn / focus on:

- provider timeout
- unknown state
- retries
- idempotency across provider
- webhook verification
- reconciliation
- DLQ
- audit

Skip / avoid in this block:

- Happy-path diagram.

Hands-on work:

- Write failure matrix: component, failure, detection, user-visible state, recovery.

Interview questions you must answer:

- Can you retry a charge?
- How reconcile mismatches?

EXIT CRITERIA: Have a payment failure matrix with at least 10 cases.

3:00 PM - 5:00 PM PostgreSQL - MVCC and isolation deeper

Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- Transactions/concurrency sections
- Index/query plan sections

What to learn / focus on:

- MVCC mental model
- READ COMMITTED / REPEATABLE READ / SERIALIZABLE
- write skew concept
- lock interaction

Skip / avoid in this block:

- Only memorizing ANSI anomalies.

Hands-on work:

- Use two psql sessions or reason through account balance reads under different isolation.

Interview questions you must answer:

- Why MVCC helps readers/writers?
- Serializable trade-off?

EXIT CRITERIA: Choose isolation for ledger update and reporting query.

6:00 PM - 8:00 PM React rendering model - deeper

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- React Essentials - Deep Dive
- behind-the-scenes / DevTools / hooks material

What to learn / focus on:

- render vs commit concept
- state batching
- component identity
- keys
- effect synchronization vs event handler

Skip / avoid in this block:

- useEffect for derivable values.

Hands-on work:

- Remove one unnecessary effect from a sample/PAT-FOS component.

Interview questions you must answer:

- When not to use useEffect?
- Why key resets state?

EXIT CRITERIA: Explain why a component re-rendered.

8:00 PM - 9:00 PM Docker - multi-stage and build optimization

Course: Docker & Kubernetes: The Practical Guide - Maximilian Schwarzmuller / Academind

Updated Apr 2026; 16 sections / 263 lectures

Course section / chapter to open:

- Docker Images & Containers
- Deploying Docker Containers -> From Development To Production

What to learn / focus on:

- layer cache
- COPY ordering
- dependency cache
- non-root runtime concept
- immutable tags

Skip / avoid in this block:

- Kubernetes today.

Hands-on work:

- Optimize Spring Boot Dockerfile for cache; tag with commit SHA concept.

Interview questions you must answer:

- Why COPY pom.xml before src?
- latest tag risk?

EXIT CRITERIA: Explain image reproducibility and rollback.

9:00 PM - 10:00 PM AI RAG - chunking and retrieval quality

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Breaking Down Big Documents - The Art of Chunking for LLMs
- RAG Made Easy - RetrievalAugmentationAdvisor in Action

What to learn / focus on:

- semantic chunk boundaries
- overlap
- topK
- threshold
- metadata filtering
- retrieval precision/recall intuition

Skip / avoid in this block:

- Agent tools.

Hands-on work:

- Evaluate 10 PAT-FOS questions: mark whether correct source chunk was retrieved.

Interview questions you must answer:

- Retrieval miss vs generation hallucination?

EXIT CRITERIA: Create retrieval evaluation sheet and tune one parameter.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 12 - BFS Level 2 - Spring proxies, topological systems and Kafka delivery

Day goal: Understand framework indirection and delivery semantics.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM **LeetCode / DSA**

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Topological Sort

Assigned problems:

- Course Schedule II
- Alien Dictionary (if available)
- Minimum Height Trees

What to learn / focus on:

- Kahn algorithm
- indegree
- DAG dependency reasoning

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM **Spring AOP / @Transactional internals**

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- Spring Core
- Spring AOP material in course
- transaction hands-on

What to learn / focus on:

- proxy-based interception
- JDK vs class proxy concept
- self-invocation pitfall
- transaction boundary
- method visibility
- AOP cross-cutting concerns

Skip / avoid in this block:

- AspectJ detail.

Hands-on work:

- Create service method A -> self-call B annotated transactional; reason/test behavior. Refactor to bean boundary.

Interview questions you must answer:

- Why @Transactional may not work on self-invocation?
- What is a proxy?

EXIT CRITERIA: Explain how annotation becomes behavior at runtime.

1:00 PM - 3:00 PM System Design - Distributed Job Scheduler

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Designing Systems that Scale -> failover/sharding
- Working with Big Data -> Message Queues
- Design Interview Strategies

What to learn / focus on:

- schedule storage
- due-time index
- scheduler leader/partitioning
- worker queue
- leases
- retry
- idempotent jobs

Skip / avoid in this block:

- Single cron server SPOF.

Hands-on work:

- Design millions of scheduled jobs with at-least-once execution.

Interview questions you must answer:

- How avoid two schedulers dispatching same job?
- Clock skew?

EXIT CRITERIA: Explain lease/idempotency based execution.

3:00 PM - 5:00 PM Kafka - delivery semantics and idempotent consumer

Course: Modern Event-Driven Spring Boot: Kafka & Spring Cloud Stream - Vinoth Selvaraj

Updated Jun 2026; Spring Boot 4 / Kafka 4; 20 sections / 190 lectures

Course section / chapter to open:

- Kafka Core Fundamentals -> offsets
- Spring Cloud Stream Consumer Applications
- failure handling topics -> retries / DLQ / error strategies

What to learn / focus on:

- at-most-once
- at-least-once
- exactly-once concept
- offset commit timing
- idempotent consumer
- duplicate event

Skip / avoid in this block:

- Marketing phrase exactly-once without scope.

Hands-on work:

- Simulate duplicate fraud assessment and ensure DB state transition is idempotent.

Interview questions you must answer:

- Crash after DB write before offset commit?
- Where store dedupe key?

EXIT CRITERIA: Explain at-least-once with a crash timeline.

6:00 PM - 8:00 PM TypeScript - generics and reusable APIs

Course: Mastering TypeScript - 2026 Edition - Colt Steele

Updated Jun 2026; 19 sections / 151 lectures

Course section / chapter to open:

- Generics
- Type Narrowing
- Type Declarations
- TypeScript Module System

What to learn / focus on:

- generic function/interface
- constraints
- keyof concept
- generic API response
- unknown errors

Skip / avoid in this block:

- Overly clever conditional types today.

Hands-on work:

- Build ApiResponse<T>, PaginatedResponse<T>, and typed fetch helper.

Interview questions you must answer:

- Generic vs union?
- Why unknown for caught errors?

EXIT CRITERIA: Use generic API types in React without any.

8:00 PM - 9:00 PM Kubernetes - probes and HPA

Course: Docker & Kubernetes: The Practical Guide - Maximilian Schwarzmuller / Academind

Updated Apr 2026; 16 sections / 263 lectures

Course section / chapter to open:

- Kubernetes resource/deployment sections
- Kubernetes core / networking; probe and scaling material

What to learn / focus on:

- readiness vs liveness
- resource requests/limits
- HPA signals
- rolling update

Skip / avoid in this block:

- EKS provisioning.

Hands-on work:

- Add /actuator/health readiness/liveness probe config to deployment YAML.

Interview questions you must answer:

- What if liveness points to DB health?
- HPA on CPU vs consumer lag?

EXIT CRITERIA: Explain safe rollout and health checks.

9:00 PM - 10:00 PM AI - reranking / advanced retrieval

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Advanced RAG with Pre-Retrieval
- Advanced RAG with Post-Retrieval

What to learn / focus on:

- query rewrite
- filtering
- reranking concept
- context compression
- hybrid retrieval concept

Skip / avoid in this block:

- MCP.

Hands-on work:

- Take 5 ambiguous questions; rewrite them into better retrieval queries and compare results.

Interview questions you must answer:

- Why rerank?
- When topK vector similarity is insufficient?

EXIT CRITERIA: Explain an advanced RAG pipeline with pre/post retrieval.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 13 - BFS Level 2 - JVM internals, Redis coordination and vector stores

Day goal: Deepen runtime and distributed cache knowledge.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM **LeetCode / DSA**

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Union Find

Assigned problems:

- Number of Connected Components
- Redundant Connection
- Accounts Merge

What to learn / focus on:

- path compression
- union by rank
- connectivity use cases

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM **JVM internals - deeper**

Course: Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

Course section / chapter to open:

- Course Specifics -> JVM Internals
- Java: A High-level Overview -> Java Virtual Machine (JVM)
- Java SE Release Cycles & Release History

What to learn / focus on:

- classloader hierarchy
- bytecode -> JIT
- heap generations concept
- GC pause vs throughput
- thread dump/heap dump use
- memory leak patterns

Skip / avoid in this block:

- Collector tuning flags.

Hands-on work:

- Create scenario: high p99 latency. List evidence to distinguish GC, DB, Kafka, CPU, thread pool.

Interview questions you must answer:

- What is JIT?
- How can Java leak memory?
- Heap vs stack?

EXIT CRITERIA: Give a performance troubleshooting decision tree.

1:00 PM - 3:00 PM System Design - Chat System

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Designing Systems that Scale -> caching / failover
- Working with Big Data -> Message Queues
- Distributed storage concepts

What to learn / focus on:

- WebSocket gateway
- presence
- message persistence
- ordering per conversation
- fanout
- offline delivery
- read receipts

Skip / avoid in this block:

- Global total ordering.

Hands-on work:

- Design 1:1 and group chat; choose partition key conversationId.

Interview questions you must answer:

- WebSocket scaling?
- Message ordering?
- Presence consistency?

EXIT CRITERIA: Separate message write path from delivery path.

3:00 PM - 5:00 PM Redis - concurrency, locks, sorted sets and Streams

Course: Redis: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; Redis 7; 22 sections / 187 lectures

Course section / chapter to open:

- Course concurrency/synchronization material
- sorted sets / lists / sets
- Streams durability material

What to learn / focus on:

- SET NX PX lock concept
- lock expiry problem
- sorted-set ranking
- Redis Streams vs Pub/Sub
- transactions/Lua concept

Skip / avoid in this block:

- Claiming Redis lock is always enough for money movement.

Hands-on work:

- Implement a leaderboard/rate limiter with sorted set; outline safe lock with token ownership.

Interview questions you must answer:

- Pub/Sub vs Streams?

- What if lock holder pauses beyond TTL?

EXIT CRITERIA: Explain when Redis coordination is appropriate and when DB locking is safer.

6:00 PM - 8:00 PM React custom hooks + reusable UI logic

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- Working with built-in Hooks and building custom Hooks
- TypeScript Generics integration

What to learn / focus on:

- hook contract
- dependency management
- data-fetch hook
- separation of view/data logic

Skip / avoid in this block:

- Custom hook for one-off trivial state.

Hands-on work:

- Build useTransactions(filters) with typed result/loading/error/refetch.

Interview questions you must answer:

- Rules of Hooks?
- Why hook names begin use?

EXIT CRITERIA: Extract reusable logic without hiding business state.

8:00 PM - 9:00 PM AWS - RDS + ElastiCache

Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- RDS, Aurora, & ElastiCache section
- AWS database differences overview

What to learn / focus on:

- Multi-AZ
- read replica
- backup
- failover
- ElastiCache Redis
- subnet/security

Skip / avoid in this block:

- Aurora internals detail.

Hands-on work:

- Map local Postgres/Redis to RDS Multi-AZ + ElastiCache; define failure behavior.

Interview questions you must answer:

- Multi-AZ vs read replica?
- Cache source of truth?

EXIT CRITERIA: Explain database HA vs read scaling.

9:00 PM - 10:00 PM AI - vector store design

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- RAG Unleashed -> Qdrant Vector Store setup
- Advanced RAG context

What to learn / focus on:

- collection/index concept
- metadata filters
- tenant separation
- embedding version
- reindex migration

Skip / avoid in this block:

- Agent orchestration.

Hands-on work:

- Design vector collection for project docs with projectId, docType, updatedAt, section.

Interview questions you must answer:

- What if embedding model changes?
- How delete stale chunks?

EXIT CRITERIA: Have a production vector-store data lifecycle plan.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 14 - BFS Level 2 - Security filters, sharding and tool calling

Day goal: Revisit security and distribution trade-offs.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Dijkstra / Weighted Graphs

Assigned problems:

- Network Delay Time
- Path With Minimum Effort
- Cheapest Flights Within K Stops

What to learn / focus on:

- priority queue relaxation
- stale heap entries
- shortest-path constraints

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Spring Security filter chain - deeper

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- Spring Boot REST API Security
- BCrypt / JDBC Authentication / role-based access
- PAT-FOS JWT filter implementation

What to learn / focus on:

- SecurityFilterChain ordering
- OncePerRequestFilter
- SecurityContext
- authentication object
- method security
- JWT signature/expiry
- 401 entry point / 403 denied

Skip / avoid in this block:

- Form-login MVC.

Hands-on work:

- Draw exact PAT-FOS request filter chain; test expired/invalid token and wrong role.

Interview questions you must answer:

- Who returns 401?
- Why method security in addition to URL rules?

EXIT CRITERIA: Explain token request path line-by-line.

1:00 PM - 3:00 PM System Design - Uber / Location

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Designing Systems that Scale -> sharding / caching
- Algorithms -> Graphs
- Working with Big Data -> Message Queues

What to learn / focus on:

- location updates
- geospatial index concept
- nearby-driver query
- matching
- trip state
- event stream

Skip / avoid in this block:

- Writing every GPS update to one SQL table synchronously.

Hands-on work:

- Design location ingestion and rider->driver matching separately.

Interview questions you must answer:

- How shard by geography?
- Hot downtown area?

EXIT CRITERIA: Explain high-write location path and durable trip path.

3:00 PM - 5:00 PM Data partitioning / sharding / replication

Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- System Design course -> Sharding Databases / NoSQL
- Postgres partitioning concepts
- MongoDB course -> replica sets and sharding concept

What to learn / focus on:

- horizontal partition
- shard key
- range vs hash
- hot shard
- replication vs sharding
- cross-shard query

Skip / avoid in this block:

- Sharding because system is “large” without throughput evidence.

Hands-on work:

- Choose shard key for payments, orders, chat and location.

Interview questions you must answer:

- merchantId vs paymentId hash?
- How reshard?

EXIT CRITERIA: Give shard-key trade-offs for four systems.

6:00 PM - 8:00 PM React error boundaries + resilient UX**Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind**

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- HTTP transitional states
- debugging React apps
- authentication/error handling material

What to learn / focus on:

- error boundary concept
- request error normalization
- retry UI
- optimistic update caution
- pending status UX

Skip / avoid in this block:

- Automatic retry of payment POST.

Hands-on work:

- Design payment UI for provider timeout: show Pending, not Failed; allow status refresh, not new charge.

Interview questions you must answer:

- Which requests safe to retry?
- How UI reflects eventual consistency?

EXIT CRITERIA: Build failure-aware payment UX flow.**8:00 PM - 9:00 PM AWS messaging - SQS/SNS/MSK positioning****Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek**

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- AWS Integration & Messaging -> SQS / SNS / Kinesis / Amazon MQ
- AWS Containers/MSK concept from architecture knowledge

What to learn / focus on:

- queue vs pub/sub
- visibility timeout
- DLQ
- Kafka/MSK vs SQS
- SNS fanout

Skip / avoid in this block:

- Every event -> Kafka.

Hands-on work:

- Choose SQS/SNS/Kafka for 5 scenarios.

Interview questions you must answer:

- SQS vs Kafka?
- Visibility timeout?

EXIT CRITERIA: Defend messaging service by requirements.**9:00 PM - 10:00 PM AI - Tool Calling****Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy**

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Tool Calling in Action - Giving LLMs the Power to Do Things
- Introduction to Tool Calling in AI
- Letting LLMs Talk to Databases - Part 1/2/3

What to learn / focus on:

- tool schema
- @Tool
- tool context
- read vs write tool safety
- model decides call then app executes

Skip / avoid in this block:

- MCP server implementation.

Hands-on work:

- Expose read-only getPaymentStatus tool; do not expose refund yet.

Interview questions you must answer:

- Prompt vs tool?
- How authorize tools?

EXIT CRITERIA: Explain secure tool-calling execution loop.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 15 - BFS Level 2 - Testing, observability and evaluation

Day goal: Make quality measurable across backend, frontend and AI.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Backtracking

Assigned problems:

- Subsets
- Combination Sum
- Word Search

What to learn / focus on:

- decision tree
- choose-explore-unchoose
- pruning

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Java/Spring Testing - deeper

Course: Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

Course section / chapter to open:

- Java In-Depth -> JUnit / IDE testing material
- Spring Boot web/service tests in your own PAT-FOS code

What to learn / focus on:

- unit vs slice vs integration vs E2E
- JUnit 5 lifecycle/assertions
- Mockito stubbing/verify
- MockMvc
- Testcontainers
- contract test concept

Skip / avoid in this block:

- Mocking everything in integration tests.

Hands-on work:

- Add/test duplicate idempotency, 403 RBAC and DB lock behavior.

Interview questions you must answer:

- What should be mocked?
- Why Testcontainers?

- Unit vs integration boundary?

EXIT CRITERIA: Create testing pyramid for PAT-FOS with one gap to fill.

1:00 PM - 3:00 PM System Design - Metrics / Logging Platform

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Working with Big Data
- Designing Systems that Scale -> distributed storage
- Cloud computing overview

What to learn / focus on:

- log ingestion
- batching
- partitioning
- retention
- index/search
- metrics time-series
- alert pipeline

Skip / avoid in this block:

- Single relational table for all raw logs.

Hands-on work:

- Design write-heavy ingestion and query path.

Interview questions you must answer:

- Backpressure?
- Retention tiers?
- Cardinality explosion?

EXIT CRITERIA: Explain write path, storage tiers and query path.

3:00 PM - 5:00 PM PostgreSQL query plans - deeper

Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- Index/query optimization material
- Subqueries / CTE material
- EXPLAIN exercises

What to learn / focus on:

- seq scan vs index scan
- cost intuition
- rows estimates
- join strategy concept
- CTE/subquery readability

Skip / avoid in this block:

- Adding index before checking plan.

Hands-on work:

- Pick slow query; collect EXPLAIN ANALYZE and propose one change.

Interview questions you must answer:

- Why index not used?
- What makes query sargable?

EXIT CRITERIA: Write before/after performance note.

6:00 PM - 8:00 PM React Testing**Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind**

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- Unit Tests
- React Testing Library concepts
- TypeScript test typing

What to learn / focus on:

- test behavior not implementation
- render/query/user event
- async wait
- mock network boundary

Skip / avoid in this block:

- Snapshot-only testing.

Hands-on work:

- Test TransferForm validation + pending status rendering.

Interview questions you must answer:

- What to mock?
- getByRole vs implementation selector?

EXIT CRITERIA: Have 2 meaningful UI tests.**8:00 PM - 9:00 PM CI - test gates and reports****Course: Jenkins: Jobs, Pipelines, CI/CD and DevOps for Beginners - Valentin Despa**

Current 2026 catalog; 7 sections / 118 lectures

Course section / chapter to open:

- Implementing Continuous Integration (CI) with Jenkins -> Running tests / Publishing JUnit report / E2E Playwright
- How to structure a pipeline?

What to learn / focus on:

- parallel test stages
- artifact/report
- fail-fast
- E2E after build
- quality gate

Skip / avoid in this block:

- Deploy before test completion.

Hands-on work:

- Update conceptual pipeline so unit tests run parallel; integration gates image; E2E gates deploy.

Interview questions you must answer:

- Why parallelize?
- What report survives failure?

EXIT CRITERIA: Explain quality gates in Jenkins and GitHub Actions.**9:00 PM - 10:00 PM AI - RAG evaluation****Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy**

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- From Testing to Production - Making AI Answers Safer with Evaluators (course later sections)
- RAG retrieval tests

What to learn / focus on:

- golden questions
- faithfulness
- answer relevance
- retrieval hit
- grounded refusal
- regression set

Skip / avoid in this block:

- Judging by “sounds good”.

Hands-on work:

- Create 10 PAT-FOS Q/A cases with expected source and pass criteria.

Interview questions you must answer:

- How distinguish retrieval vs generation error?

EXIT CRITERIA: Have a repeatable AI evaluation set.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 16 - BFS Level 2 - Microservice communication, file systems and frontend architecture

Day goal: Compare sync/async boundaries and design large-object storage.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM **LeetCode / DSA**

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Trie

Assigned problems:

- Implement Trie
- Design Add and Search Words
- Word Search II

What to learn / focus on:

- prefix tree
- node branching
- space trade-offs

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM **Microservice communication - REST / Feign / events**

Course: [NEW] Master Microservices with SpringBoot,Docker,Kubernetes - Eazy Bytes / Madan Reddy

Updated Aug 2026; 21 sections / 282 lectures

Course section / chapter to open:

- Service Discovery & Registration with Eureka
- Feign Client code changes
- event-driven microservices section (RabbitMQ/Kafka/Spring Cloud Stream)

What to learn / focus on:

- sync request/response
- service discovery
- client-side load balance
- async events
- temporal coupling
- contract versioning

Skip / avoid in this block:

- Microservice for every class/module.

Hands-on work:

- Map PAT-FOS fraud communication: compare current Kafka async vs hypothetical Feign synchronous call.

Interview questions you must answer:

- When REST?
- When Kafka?
- What is temporal coupling?

EXIT CRITERIA: Choose sync vs async for five interactions.

1:00 PM - 3:00 PM System Design - File Storage / Dropbox

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Designing Systems that Scale -> Distributed Storage Solutions / HDFS architecture
- CDN
- Mock design interview patterns

What to learn / focus on:

- metadata vs blob
- S3/object storage
- chunked upload
- dedupe/hash
- versioning
- sync events
- CDN

Skip / avoid in this block:

- Storing large file bytes in application DB rows.

Hands-on work:

- Design upload/download; use presigned URL concept and metadata DB.

Interview questions you must answer:

- Large upload resume?
- Consistency of metadata vs object?

EXIT CRITERIA: Explain control plane vs data plane.

3:00 PM - 5:00 PM MongoDB - aggregation + sharding

Course: MongoDB - The Complete Developer's Guide - Maximilian Schwarzmuller / Academind

Updated Jan 2026; 19 sections / 266 lectures

Course section / chapter to open:

- Aggregation Framework material
- Working with Indexes
- replica sets and sharding modules

What to learn / focus on:

- \$match/\$group/\$project/\$sort concept
- pipeline order
- index-before-aggregation
- replica set
- shard key

Skip / avoid in this block:

- Every analytics job in app memory.

Hands-on work:

- Aggregate fraud assessments by reason/decision/day. Propose shard key for huge assessment collection.

Interview questions you must answer:

- \$match early why?
- Replica vs shard?

EXIT CRITERIA: Build one useful aggregation and explain scale path.

6:00 PM - 8:00 PM Frontend architecture - feature boundaries

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- React Essentials project structure/best practices
- Context/Redux/routing/HTTP sections recap
- Combining React with TypeScript

What to learn / focus on:

- feature folders
- UI vs API/domain types
- hooks/services/components
- server state vs client state concept

Skip / avoid in this block:

- One giant components folder.

Hands-on work:

- Refactor PAT-FOS feature boundaries on paper/code: auth, accounts, transfers, admin.

Interview questions you must answer:

- Where should API DTO live?
- Where business transform?

EXIT CRITERIA: Produce maintainable frontend folder map.

8:00 PM - 9:00 PM AWS S3 + CloudFront

Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- S3 sections
- CloudFront / CDN sections

What to learn / focus on:

- object key
- bucket policy
- presigned URL
- versioning
- CDN cache
- origin

Skip / avoid in this block:

- Public S3 bucket by default.

Hands-on work:

- Design React static hosting and document upload via S3/CloudFront.

Interview questions you must answer:

- Presigned URL benefit?
- Cache invalidation?

EXIT CRITERIA: Explain secure direct-to-S3 upload.

9:00 PM - 10:00 PM AI - agent concept

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Tool Calling section recap
- Capstone Project - Building a Real-World AI Agent (preview)

What to learn / focus on:

- agent loop
- planning vs deterministic workflow
- tools
- memory
- stopping conditions
- human approval

Skip / avoid in this block:

- Calling every tool automatically.

Hands-on work:

- Design payment-support agent that can read payment but requires approval for refund.

Interview questions you must answer:

- Agent vs workflow?
- Where human-in-the-loop?

EXIT CRITERIA: Have safe agent capability matrix.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 17 - BFS Level 2 - Saga, outbox and production state

Day goal: Master consistency patterns across services.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Greedy

Assigned problems:

- Jump Game
- Gas Station
- Partition Labels

What to learn / focus on:

- local invariant
- proof idea
- when greedy fails

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Distributed transactions - Saga patterns

Course: [NEW] Master Microservices with SpringBoot,Docker,Kubernetes - Eazy Bytes / Madan Reddy

Updated Aug 2026; 21 sections / 282 lectures

Course section / chapter to open:

- Event-driven microservices section
- Resilience4J section as failure context
- API Gateway patterns if orchestration endpoint needed

What to learn / focus on:

- local transaction
- saga choreography
- saga orchestration
- compensating action
- idempotent step
- state machine

Skip / avoid in this block:

- 2PC as default across internet-scale services.

Hands-on work:

- Model Order -> Reserve Inventory -> Charge Payment -> Create Shipment and compensations.

Interview questions you must answer:

- Compensation if payment charged but inventory later fails?

EXIT CRITERIA: Draw saga state and compensation table.

1:00 PM - 3:00 PM System Design - E-commerce Checkout

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Design Interview Strategies
- Message Queues
- Caching / resiliency / ACID-CAP

What to learn / focus on:

- cart
- pricing
- inventory
- order
- payment
- fraud
- promotion
- notification
- idempotency

Skip / avoid in this block:

- One huge checkout transaction across microservices.

Hands-on work:

- Design critical synchronous steps vs async aftermath.

Interview questions you must answer:

- Price changes?
- Inventory race?
- Payment unknown?

EXIT CRITERIA: Explain checkout consistency boundaries.

3:00 PM - 5:00 PM Transactional Outbox

Course: Modern Event-Driven Spring Boot: Kafka & Spring Cloud Stream - Vinoth Selvaraj

Updated Jun 2026; Spring Boot 4 / Kafka 4; 20 sections / 190 lectures

Course section / chapter to open:

- Kafka failure strategies + producer/consumer concepts
- Spring event-driven architecture
- apply database pattern in PAT-FOS

What to learn / focus on:

- dual-write problem
- outbox row in same DB transaction
- publisher polling/CDC
- at-least-once event
- consumer idempotency

Skip / avoid in this block:

- Publish to Kafka inside DB transaction and assume atomicity.

Hands-on work:

- Design outbox_events schema and publisher; trace crash at every point.

Interview questions you must answer:

- Publisher crash after send before mark sent?
- How clean outbox?

EXIT CRITERIA: Explain why outbox solves DB+Kafka atomic intent, not duplicates.

6:00 PM - 8:00 PM Frontend state trade-offs

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- Context API
- Redux & Redux Toolkit
- HTTP / forms sections

What to learn / focus on:

- local state
- context
- Redux
- URL state
- server-state cache concept
- avoid duplication

Skip / avoid in this block:

- Putting every API response into Redux.

Hands-on work:

- Classify 10 PAT-FOS state values by best location.

Interview questions you must answer:

- When Redux?
- When URL?
- Why duplicate derived state dangerous?

EXIT CRITERIA: Create state ownership matrix.

8:00 PM - 9:00 PM Observability - metrics/logs/traces

Course: [NEW] Master Microservices with SpringBoot,Docker,Kubernetes - Eazy Bytes / Madan Reddy

Updated Aug 2026; 21 sections / 282 lectures

Course section / chapter to open:

- Implementing observability and monitoring using Prometheus, Loki, Promtail, Tempo and Grafana

What to learn / focus on:

- RED metrics
- structured log
- correlation id
- trace/span
- Kafka lag
- SLO alert

Skip / avoid in this block:

- Logging everything at ERROR.

Hands-on work:

- Define dashboard for payment API: p95/p99, 5xx, pending count, Kafka lag, DB pool.

Interview questions you must answer:

- Metric vs log vs trace?

EXIT CRITERIA: Have monitoring/alerting plan.

9:00 PM - 10:00 PM AI - memory

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Adding Memory to LLM Conversations
- Spring AI Memory Advisors
- JDBC-Based Persistence
- maxMessages / context window

What to learn / focus on:

- conversationId
- short-term window
- persistent chat repository
- token budget
- memory vs RAG

Skip / avoid in this block:

- Treat chat history as trusted factual source.

Hands-on work:

- Design per-user support-chat memory with max messages and DB persistence.

Interview questions you must answer:

- Memory vs RAG?
- Cross-user leakage prevention?

EXIT CRITERIA: Explain secure multi-user memory.**End-of-day checkpoint**

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 18 - BFS Level 2 - Virtual threads, distributed cache and CAP

Day goal: Understand modern Java concurrency and consistency choices.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM **LeetCode / DSA**

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Dynamic Programming - foundations

Assigned problems:

- Climbing Stairs
- House Robber
- Coin Change

What to learn / focus on:

- state definition
- transition
- top-down vs bottom-up

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM **Virtual Threads - deeper**

Course: Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

Course section / chapter to open:

- Modern Java Features -> Virtual Threads
- Concurrency section (including Virtual Threads)

What to learn / focus on:

- carrier thread concept
- blocking I/O unmount
- structured concurrency concept
- pinning caveat
- connection pool remains bottleneck

Skip / avoid in this block:

- Claiming millions of DB calls because millions of virtual threads.

Hands-on work:

- Compare fixed pool vs virtual-thread executor for simulated sleeping I/O; note DB pool limits.

Interview questions you must answer:

- CPU-bound benefit?
- What resource still limits?

EXIT CRITERIA: Explain where virtual threads help Spring Boot and where they do not.

1:00 PM - 3:00 PM System Design - Distributed Cache**Course: Mastering the System Design Interview - Frank Kane / Sundog Education**

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Caching + Eviction Strategies
- Sharding Databases / NoSQL
- Resiliency

What to learn / focus on:

- consistent hashing concept
- replication
- TTL
- eviction
- hot key
- cache stampede
- failover

Skip / avoid in this block:

- One Redis node.

Hands-on work:

- Design distributed cache get/put and node-add/remove behavior.

Interview questions you must answer:

- Hot key?
- Cold start?
- Consistency with DB?

EXIT CRITERIA: Explain cache partition/replication trade-off.**3:00 PM - 5:00 PM CAP / PACELC and consistency****Course: Mastering the System Design Interview - Frank Kane / Sundog Education**

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- ACID compliance and the CAP theorem
- Using CAP to Choose a Database

What to learn / focus on:

- partition tolerance
- CP/AP response during partition
- eventual consistency
- PACELC latency-consistency trade-off
- quorum intuition

Skip / avoid in this block:

- Saying “choose any 2” without noting partitions are unavoidable.

Hands-on work:

- Classify payment ledger, cache, feed, notification, search indexes.

Interview questions you must answer:

- Is PostgreSQL simply CP? nuance
- Where stale reads acceptable?

EXIT CRITERIA: Use consistency requirements to justify technology, not labels.**6:00 PM - 8:00 PM React performance profiling**

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- React DevTools
- behind-the-scenes rendering
- Hooks performance material

What to learn / focus on:

- Profiler
- expensive render
- memoization
- component splitting
- virtualization concept

Skip / avoid in this block:

- Memo everything.

Hands-on work:

- Profile transaction table and note top render cost; optimize one measurable issue.

Interview questions you must answer:

- When memo hurts?

EXIT CRITERIA: Performance change backed by evidence.**8:00 PM - 9:00 PM AWS CloudWatch / operational basics****Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek**

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- Monitoring, Troubleshooting & Audit -> CloudWatch / CloudTrail / EventBridge / Config

What to learn / focus on:

- metrics
- logs
- alarms
- CloudTrail audit
- EventBridge events

Skip / avoid in this block:

- Alert on every single error log.

Hands-on work:

- Map PAT-FOS log/metrics to CloudWatch and define 4 alarms.

Interview questions you must answer:

- CloudWatch vs CloudTrail?

EXIT CRITERIA: Have production observability mapping.**9:00 PM - 10:00 PM AI - MCP first pass****Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy**

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- MCP section -> Demo of MCP Client and MCP Server integration
- Meet the MCP Inspector
- Meet the MCP Annotation Family (preview)

What to learn / focus on:

- client/server/tool/resource/prompt

- transport concept
- tool discovery
- MCP vs direct tool integration

Skip / avoid in this block:

- Building remote server today.

Hands-on work:

- List PAT-FOS tools an MCP server could expose: get-payment, get-fraud-assessment, search-docs.

Interview questions you must answer:

- MCP vs REST?
- Why standardize tool schema?

EXIT CRITERIA: Explain MCP in one architecture diagram.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 19 - BFS Level 2 - Performance, hot partitions and production AI safety

Day goal: Recognize bottlenecks and design for degraded conditions.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Dynamic Programming - medium

Assigned problems:

- Longest Increasing Subsequence
- Word Break
- Partition Equal Subset Sum

What to learn / focus on:

- 1D/2D state
- optimization
- recognizing repeated subproblems

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Spring Boot performance

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- Spring Boot / JPA sections + hands-on profiling
- Java JVM internals reference

What to learn / focus on:

- Hikari pool
- DB slow query
- N+1
- HTTP client pool
- serialization
- cache hit rate
- thread saturation
- p99

Skip / avoid in this block:

- Tuning heap before measuring.

Hands-on work:

- Create incident scenario: p99 2s, CPU 25%. Work through DB pool, downstream latency, Kafka lag.

Interview questions you must answer:

- How detect connection pool exhaustion?
- Why average latency insufficient?

EXIT CRITERIA: Give a production troubleshooting sequence.

1:00 PM - 3:00 PM System Design - Notification System v2

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Message Queues
- Resiliency
- Caching / failover

What to learn / focus on:

- backpressure
- provider quota
- priority queues
- preference store
- dedupe
- multi-region
- DLQ replay

Skip / avoid in this block:

- Same design as Day3 without improvements.

Hands-on work:

- Redo under 1M notifications/min and provider outage.

Interview questions you must answer:

- How prioritize OTP over marketing?
- How replay DLQ?

EXIT CRITERIA: Produce v2 architecture with backpressure.

3:00 PM - 5:00 PM Data failure patterns

Course: Redis: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; Redis 7; 22 sections / 187 lectures

Course section / chapter to open:

- Redis concepts + Kafka + Postgres synthesis

What to learn / focus on:

- hot partition
- cache stampede
- cache penetration
- replication lag
- thundering herd
- consumer lag

Skip / avoid in this block:

- One generic “add Redis” fix.

Hands-on work:

- For each failure, write symptom -> metric -> mitigation.

Interview questions you must answer:

- How prevent stampede?
- What does replica lag break?

EXIT CRITERIA: Create six-pattern failure cheat sheet.

6:00 PM - 8:00 PM React real-time status UI

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- HTTP transitional states
- side effects/custom hooks
- routing/state recap

What to learn / focus on:

- polling vs SSE vs WebSocket
- backoff
- visibility pause
- eventual state

Skip / avoid in this block:

- Poll every 100ms forever.

Hands-on work:

- Improve payment polling: exponential interval/stop terminal state; sketch SSE alternative.

Interview questions you must answer:

- Polling vs SSE?
- What happens tab hidden?

EXIT CRITERIA: Build/describe scalable status update UX.

8:00 PM - 9:00 PM Blue-green / canary deployment

Course: Jenkins: Jobs, Pipelines, CI/CD and DevOps for Beginners - Valentin Despa

Current 2026 catalog; 7 sections / 118 lectures

Course section / chapter to open:

- Continuous Delivery / deploying to production sections
- GitHub Actions environment / workflow best practices

What to learn / focus on:

- immutable artifact
- blue-green
- canary
- health gate
- rollback
- DB migration compatibility

Skip / avoid in this block:

- Deploy app and destructive migration simultaneously.

Hands-on work:

- Design zero-downtime deploy of Spring API with additive DB migration.

Interview questions you must answer:

- Backward-compatible migration?
- Rollback after schema change?

EXIT CRITERIA: Explain safe release sequence.

9:00 PM - 10:00 PM AI production safety

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- From Testing to Production - safer answers with Evaluators
- Observability in Spring AI

What to learn / focus on:

- prompt injection
- tool authorization
- PII
- grounding
- output validation
- cost/token monitoring
- fallback

Skip / avoid in this block:

- Let model decide authorization.

Hands-on work:

- Threat-model payment AI assistant; define read/write tool policies.

Interview questions you must answer:

- How stop prompt injection from docs?
- What logs safe?

EXIT CRITERIA: Create AI production safety checklist.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 20 - BFS Pass 2 checkpoint - interview proof

Day goal: Measure Level-2 depth and decide where the final week goes.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: BFS Pass-2 Coding Mock

Assigned problems:

- 2 timed mediums + 1 follow-up

What to learn / focus on:

- 45-minute discipline
- edge cases
- clean explanation

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Java/Spring interview simulation

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course except to patch misses after simulation

What to learn / focus on:

- concurrency internals
- JVM
- Spring proxy/transactions
- security
- testing
- performance

Skip / avoid in this block:

- Notes during first attempt.

Hands-on work:

- Run 12-question mock; answer each in 2-4 minutes. Then review misses.

Interview questions you must answer:

- Follow-up aggressively: why/how/failure/trade-off.

EXIT CRITERIA: Score /10 per category and update RED/YELLOW/GREEN.

1:00 PM - 3:00 PM System Design - unknown mock

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Use Mastering System Design only after mock

What to learn / focus on:

- framework execution under uncertainty
- trade-offs
- failure handling

Skip / avoid in this block:

- Known memorized design.

Hands-on work:

- Choose randomly: Job Scheduler / Chat / Logging / Reservation. 45-min mock + debrief.

Interview questions you must answer:

- Why each DB?
- What fails?
- 10x traffic?

EXIT CRITERIA: Complete design within 45 minutes.

3:00 PM - 5:00 PM Data/Kafka interview simulation

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course first 90 min

What to learn / focus on:

- indexes
- transactions/MVCC
- locks
- Redis
- Kafka semantics
- outbox
- sharding

Skip / avoid in this block:

- Definitions only.

Hands-on work:

- Run 15 rapid questions and 3 scenarios.

Interview questions you must answer:

- Crash timelines and duplicate handling.

EXIT CRITERIA: Identify top three data weaknesses.

6:00 PM - 8:00 PM React/TypeScript interview simulation

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course first

What to learn / focus on:

- hooks
- rendering
- state ownership
- typing
- routing/auth
- testing/performance

Skip / avoid in this block:

- Coding by copy.

Hands-on work:

- Build one typed searchable transaction list while explaining choices.

Interview questions you must answer:

- Why useEffect? why not?
- Context vs Redux?

EXIT CRITERIA: Frontend scorecard.

8:00 PM - 9:00 PM AWS/DevOps mock

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Reference after mock

What to learn / focus on:

- Docker
- K8s
- CI/CD
- RDS/Redis/ECS
- observability

Skip / avoid in this block:

- Service-definition memorization.

Hands-on work:

- Explain local -> AWS deployment and one failure rollback.

Interview questions you must answer:

- Multi-AZ? secrets?

EXIT CRITERIA: Cloud/DevOps scorecard.

9:00 PM - 10:00 PM AI mock

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Spring AI course only after

What to learn / focus on:

- RAG
- embeddings
- tools
- memory
- MCP
- evaluation/safety

Skip / avoid in this block:

- Buzzwords.

Hands-on work:

- Explain one AI feature end-to-end and defend safety.

Interview questions you must answer:

- RAG vs fine-tune? tools vs MCP?

EXIT CRITERIA: AI scorecard and Pass-3 priorities.

End-of-day checkpoint

- Write 3 things you can now explain without notes.

- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 21 - BFS Level 3 - Production Payment Architecture

Day goal: Integrate backend, data, frontend, cloud and AI around one payment design.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM **LeetCode / DSA**

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Timed interview - arrays/graphs

Assigned problems:

- 1 random medium in 45 min + follow-up

What to learn / focus on:

- clarify
- design
- code
- test
- complexity

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM **Backend payment implementation patterns**

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- Spring transactions/security/JPA review
- Microservices Resilience4j
- Kafka/outbox prior sections

What to learn / focus on:

- idempotency
- state machine
- provider adapter
- transaction boundary
- pessimistic lock
- outbox
- audit

Skip / avoid in this block:

- New beginner content.

Hands-on work:

- Write/refactor PaymentService pseudocode or PAT-FOS service into explicit phases: create -> fraud -> settle -> publish.

Interview questions you must answer:

- Where transaction starts/ends?
- Which call outside DB transaction?

EXIT CRITERIA: A code-level architecture matching system design.

1:00 PM - 3:00 PM System Design - Payment System v3 production

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Design Interview Strategies / Resiliency / CAP / Message Queues recap

What to learn / focus on:

- ledger
- payment attempts
- webhooks
- reconciliation
- outbox
- multi-provider routing
- fraud
- observability
- multi-region

Skip / avoid in this block:

- Simple flow only.

Hands-on work:

- 45-min design + 45-min failure challenges + 30-min rewrite.

Interview questions you must answer:

- Exactly-once money?
- provider failover can double charge?
- home region?

EXIT CRITERIA: Final payment design you can present in interview.

3:00 PM - 5:00 PM Payment data correctness

Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- Transactions/locks/indexes + Kafka idempotency synthesis

What to learn / focus on:

- ledger schema
- unique idempotency
- payment_attempt
- webhook_event dedupe
- reconciliation query
- indexes

Skip / avoid in this block:

- Status field as only financial truth.

Hands-on work:

- Draft schema and indexes; define reconciliation query for PENDING older than N minutes.

Interview questions you must answer:

- Ledger vs payment status?
- Partial refund?

EXIT CRITERIA: Production payment schema.

6:00 PM - 8:00 PM Payment frontend UX**Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind**

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- HTTP state / forms / auth / TypeScript unions

What to learn / focus on:

- disable duplicate submit
- idempotency key lifecycle
- pending UX
- status polling/SSE
- retry status not charge

Skip / avoid in this block:

- Generate new idempotency key on network retry of same user intent.

Hands-on work:

- Build flow diagram and update TransferForm behavior.

Interview questions you must answer:

- When reuse idempotency key?

EXIT CRITERIA: Frontend prevents accidental duplicate intent and handles uncertainty.**8:00 PM - 9:00 PM Payment deployment/observability****Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek**

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- ECS/RDS/ElastiCache/CloudWatch sections
- microservice observability recap

What to learn / focus on:

- ALB
- ECS Fargate
- RDS Multi-AZ
- Redis
- MSK
- CloudWatch
- Secrets

Skip / avoid in this block:

- Single-AZ prod.

Hands-on work:

- Draw AWS payment architecture and 5 alarms.

Interview questions you must answer:

- What fails if AZ gone?

EXIT CRITERIA: Production cloud diagram.**9:00 PM - 10:00 PM AI payment investigation assistant****Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy**

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Tool Calling + RAG sections

What to learn / focus on:

- RAG docs + read-only tools
- fraud explanation
- authorization
- audit of AI action

Skip / avoid in this block:

- Automated refund.

Hands-on work:

- Design “why was payment rejected?” assistant with getPayment/getFraud tools.

Interview questions you must answer:

- Grounded answer? permissions?

EXIT CRITERIA: Safe AI extension to payment system.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 22 - BFS Level 3 - E-commerce architecture

Day goal: Integrate catalog, inventory, order, payment and fulfillment.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Timed interview - tree/heap

Assigned problems:

- 1 random medium in 45 min + follow-up

What to learn / focus on:

- communication under time pressure

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Microservice boundaries + API Gateway

Course: [NEW] Master Microservices with SpringBoot,Docker,Kubernetes - Eazy Bytes / Madan Reddy

Updated Aug 2026; 21 sections / 282 lectures

Course section / chapter to open:

- Spring Cloud support for Service Discovery & Registration
- Building Edge Server using Spring Cloud Gateway
- Design patterns around API Gateway

What to learn / focus on:

- bounded responsibility
- gateway routing
- correlation ID
- aggregation/BFF
- service discovery
- avoid shared DB

Skip / avoid in this block:

- Gateway business logic.

Hands-on work:

- Map Catalog/Cart/Inventory/Order/Payment services and ownership.

Interview questions you must answer:

- When gateway aggregation?
- Why database per service?

EXIT CRITERIA: Service boundary diagram.

1:00 PM - 3:00 PM System Design - E-commerce platform**Course: Mastering the System Design Interview - Frank Kane / Sundog Education**

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Scaling/caching/message queues/cloud recap

What to learn / focus on:

- catalog/search/cart/inventory/order/payment/notification
- read vs write paths
- sale spike

Skip / avoid in this block:

- One DB for everything.

Hands-on work:

- Design normal path then flash-sale traffic.

Interview questions you must answer:

- Inventory consistency?
- hot product?

EXIT CRITERIA: Explain 10x sale-day scaling.**3:00 PM - 5:00 PM Polyglot data choices****Course: SQL and PostgreSQL: The Complete Developer's Guide - Stephen Grider**

Updated Feb 2026; 37 sections / 289 lectures

Course section / chapter to open:

- Postgres + Redis + Kafka + Mongo synthesis

What to learn / focus on:

- catalog store
- cart Redis
- inventory SQL
- search index
- order SQL
- events Kafka

Skip / avoid in this block:

- Technology by popularity.

Hands-on work:

- Create data-ownership table: entity, source of truth, cache/index/event.

Interview questions you must answer:

- Why search index not source of truth?

EXIT CRITERIA: Data ownership matrix.**6:00 PM - 8:00 PM E-commerce React architecture****Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind**

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- routing/state/HTTP/forms/performance recap

What to learn / focus on:

- catalog pagination
- cart local/server state
- checkout state machine

- optimistic UI boundaries

Skip / avoid in this block:

- Optimistic payment success.

Hands-on work:

- Sketch feature architecture and checkout steps.

Interview questions you must answer:

- What can be optimistic?

EXIT CRITERIA: Frontend domain boundary map.

8:00 PM - 9:00 PM Kubernetes microservices deployment

Course: Docker & Kubernetes: The Practical Guide - Maximilian Schwarzmuller / Academind

Updated Apr 2026; 16 sections / 263 lectures

Course section / chapter to open:

- Kubernetes deployments/services/networking
- Helm role preview from microservices course

What to learn / focus on:

- per-service deployment
- service discovery
- ConfigMap/Secret
- rolling update

Skip / avoid in this block:

- One pod with all microservices.

Hands-on work:

- Write deployment strategy for 5 services; identify independent scaling metric.

Interview questions you must answer:

- Scale fraud by CPU or lag?

EXIT CRITERIA: Orchestration scale plan.

9:00 PM - 10:00 PM AI shopping assistant architecture

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- RAG + Tool Calling

What to learn / focus on:

- catalog RAG/search
- inventory/price real-time tools
- recommendation vs factual tools

Skip / avoid in this block:

- Let RAG answer live price from stale docs.

Hands-on work:

- Design product assistant: semantic search + getLivePrice tool.

Interview questions you must answer:

- What data via RAG vs tool?

EXIT CRITERIA: Hybrid retrieval + tool architecture.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 23 - BFS Level 3 - High-scale backend and CI automation

Day goal: Reason about 100K QPS and operational gates.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Timed interview - graph/DP

Assigned problems:

- 1 random medium in 45 min + follow-up

What to learn / focus on:

- pattern selection and trade-offs

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM High-throughput Spring backend

Course: Java In-Depth: Become a Complete Java Engineer! [2026] - Dheeru Mundluru

Updated Aug 2026; Java 25; 30 sections / 470 lectures

Course section / chapter to open:

- Concurrency/Virtual Threads/JVM + Spring performance review

What to learn / focus on:

- stateless scale
- connection pool
- backpressure
- batching
- async boundary
- virtual threads vs downstream limits

Skip / avoid in this block:

- Thread count as only scaling lever.

Hands-on work:

- Capacity worksheet: 100K QPS, DB pool, external calls, cache hit rate.

Interview questions you must answer:

- Where bottleneck first?

EXIT CRITERIA: Capacity model with assumptions.

1:00 PM - 3:00 PM System Design - 100K QPS API

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Horizontal vs Vertical Scaling
- Failover
- Caching
- Sharding
- Resiliency

What to learn / focus on:

- load balancer
- stateless servers
- cache
- partition
- queue
- rate limit
- degrade

Skip / avoid in this block:

- Premature multi-region.

Hands-on work:

- Take order lookup API and scale from 100 to 100K QPS.

Interview questions you must answer:

- What changes first? last?

EXIT CRITERIA: Incremental scaling story.

3:00 PM - 5:00 PM Backpressure + Kafka consumer scaling

Course: Modern Event-Driven Spring Boot: Kafka & Spring Cloud Stream - Vinoth Selvaraj

Updated Jun 2026; Spring Boot 4 / Kafka 4; 20 sections / 190 lectures

Course section / chapter to open:

- Consumer Group Scaling Scenarios
- Partition Rebalancing
- parallel/batch processing topics

What to learn / focus on:

- partition count
- consumer replicas
- lag
- batch size
- pause/resume concept
- downstream DB pressure

Skip / avoid in this block:

- Add consumers beyond partitions.

Hands-on work:

- Model fraud lag with 12 partitions, variable service latency.

Interview questions you must answer:

- How auto-scale on lag?
- What if DB cannot keep up?

EXIT CRITERIA: End-to-end backpressure plan.

6:00 PM - 8:00 PM Frontend large-list performance

Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- React performance/DevTools
- HTTP pagination
- React Router data concepts

What to learn / focus on:

- cursor pagination
- virtualization concept
- debounce search
- request cancellation

Skip / avoid in this block:

- Load 100k rows into browser.

Hands-on work:

- Implement/outline cursor transaction list and debounced filter.

Interview questions you must answer:

- Offset vs cursor?

EXIT CRITERIA: Scalable data table design.

8:00 PM - 9:00 PM GitHub Actions - advanced/reusable

Course: GitHub Actions: The Complete Guide from Beginner to Expert - Lauro Fialho Muller

Updated Aug 2026; 20 sections / 115 lectures

Course section / chapter to open:

- Advanced concepts -> custom actions and reusable workflows
- workflow security
- triggers/schedules
- caches/artifacts

What to learn / focus on:

- reusable workflow
- matrix
- environment approval
- permissions
- OIDC concept
- cache

Skip / avoid in this block:

- Long-lived AWS keys in repo secret if OIDC available.

Hands-on work:

- Refactor conceptual PAT-FOS test jobs into reusable workflow; set least permissions.

Interview questions you must answer:

- Cache vs artifact?
- Reusable workflow vs composite action?

EXIT CRITERIA: Modern secure CI design.

9:00 PM - 10:00 PM AI latency/cost architecture

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Observability in Spring AI
- RAG / context-window material

What to learn / focus on:

- token budget
- model routing
- cache safe prompts
- streaming response
- timeouts
- fallback model

Skip / avoid in this block:

- Largest model for every request.

Hands-on work:

- Define SLO/cost budget for AI endpoint and fallback path.

Interview questions you must answer:

- How monitor tokens/cost?

EXIT CRITERIA: AI performance budget.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 24 - BFS Level 3 - Failure Day

Day goal: Break every subsystem and practice recovery.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Failure-day coding

Assigned problems:

- Redo two previously failed problems from memory

What to learn / focus on:

- error log
- root cause of wrong approach

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Backend incident response

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Use Spring/JVM/Resilience notes after diagnosis

What to learn / focus on:

- DB pool exhaustion
- downstream timeout
- thread saturation
- OOM/GC
- bad deploy

Skip / avoid in this block:

- Random code changes before evidence.

Hands-on work:

- Run 3 tabletop incidents: symptom -> metrics -> logs -> hypothesis -> mitigation -> root fix.

Interview questions you must answer:

- Immediate mitigation vs permanent fix?

EXIT CRITERIA: Incident runbook skeleton.

1:00 PM - 3:00 PM System Design - failure injection mock

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course first

What to learn / focus on:

- DB unavailable
- Redis unavailable
- Kafka unavailable
- provider timeout
- region outage

Skip / avoid in this block:

- Redraw happy path.

Hands-on work:

- Take payment/order design and inject failures one by one.

Interview questions you must answer:

- What state is user shown?
- How recover?

EXIT CRITERIA: Recovery matrix.

3:00 PM - 5:00 PM Data failure day

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Postgres/Redis/Kafka references after reasoning

What to learn / focus on:

- deadlock
- replica lag
- cache outage
- hot key
- consumer poison message
- duplicate event

Skip / avoid in this block:

- Silent drop.

Hands-on work:

- For each: detection metric, safety impact, retry/DLQ/reconciliation.

Interview questions you must answer:

- Can retry worsen issue?

EXIT CRITERIA: Operational cheat sheet.

6:00 PM - 8:00 PM Frontend failure UX

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- React HTTP/auth material

What to learn / focus on:

- offline
- timeout
- 401
- 500
- pending async operation
- partial page failure

Skip / avoid in this block:

- Generic “Something went wrong” for all states.

Hands-on work:

- Design UI copy/actions for each failure.

Interview questions you must answer:

- Which failures retryable?

EXIT CRITERIA: User-facing failure-state matrix.

8:00 PM - 9:00 PM Deployment failure / rollback

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Jenkins/GitHub Actions/K8s notes

What to learn / focus on:

- failed readiness
- bad migration
- config error
- canary rollback

Skip / avoid in this block:

- Force rollout despite health failure.

Hands-on work:

- Tabletop: new version fails readiness; explain automatic/manual rollback.

Interview questions you must answer:

- What if migration irreversible?

EXIT CRITERIA: Release rollback runbook.

9:00 PM - 10:00 PM AI failure day

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- AI evaluation/safety/RAG sections

What to learn / focus on:

- provider outage
- retrieval miss
- hallucination
- tool failure
- prompt injection
- token limit

Skip / avoid in this block:

- Trust model answer if tool failed.

Hands-on work:

- Define fallback/refusal for each case.

Interview questions you must answer:

- How surface uncertainty?

EXIT CRITERIA: AI degradation policy.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 25 - BFS Level 3 - AWS production architecture

Day goal: Move the full platform from laptop to resilient AWS deployment.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Timed interview - mixed

Assigned problems:

- 2 random medium problems

What to learn / focus on:

- speed + correctness

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Application production readiness

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- Spring Boot Actuator/security/config/JPA recap
- Microservice observability/resilience

What to learn / focus on:

- externalized config
- health probes
- graceful shutdown
- timeouts
- connection pool
- structured logging
- secrets

Skip / avoid in this block:

- Dev defaults in prod.

Hands-on work:

- Create production-readiness checklist for transaction-service.

Interview questions you must answer:

- What changes from local?

EXIT CRITERIA: Backend production checklist.

1:00 PM - 3:00 PM System Design - Multi-AZ AWS payment system

Course: Mastering the System Design Interview - Frank Kane / Sundog Education

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Cloud Computing overview
- Resiliency
- Distributed Storage
- CAP

What to learn / focus on:

- Route53/CloudFront/ALB
- ECS multi-AZ
- RDS Multi-AZ
- ElastiCache
- MSK/SQS
- S3
- CloudWatch

Skip / avoid in this block:

- Single public subnet.

Hands-on work:

- Draw exact AWS architecture with private subnets and failure domains.

Interview questions you must answer:

- AZ failure? region failure?

EXIT CRITERIA: Cloud production architecture.**3:00 PM - 5:00 PM AWS data services decision****Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek**

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- RDS, Aurora, & ElastiCache
- AWS Integration & Messaging
- AWS Containers Services
- S3

What to learn / focus on:

- RDS vs DynamoDB
- ElastiCache
- SQS/SNS vs MSK
- S3
- backup/DR

Skip / avoid in this block:

- One AWS service per buzzword.

Hands-on work:

- Map every local PAT-FOS container to managed AWS equivalent and state why.

Interview questions you must answer:

- DocumentDB vs MongoDB Atlas?
- MSK vs SQS?

EXIT CRITERIA: Local -> AWS mapping with trade-offs.**6:00 PM - 8:00 PM Frontend production hosting****Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind**

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- Deploying React Apps
- authentication / HTTP recap
- AWS CloudFront/S3 concepts

What to learn / focus on:

- static build
- S3 origin
- CloudFront
- environment config
- API base URL
- CORS

Skip / avoid in this block:

- Secrets in VITE env vars.

Hands-on work:

- Define build-time public config vs server-side secrets.

Interview questions you must answer:

- What is exposed in browser bundle?

EXIT CRITERIA: Secure frontend deployment plan.

8:00 PM - 9:00 PM AWS Secrets + monitoring

Course: Ultimate AWS Certified Solutions Architect Associate 2026 - Stephane Maarek

Updated Aug 2026; 33 sections / 397 lectures

Course section / chapter to open:

- IAM/security sections
- Monitoring, Troubleshooting & Audit
- Secrets Manager/KMS concepts in course

What to learn / focus on:

- Secrets Manager
- task role
- KMS
- CloudWatch alarm
- CloudTrail

Skip / avoid in this block:

- Static AWS keys in container.

Hands-on work:

- Define task role permissions and secret injection.

Interview questions you must answer:

- Secret rotation?

EXIT CRITERIA: Least-privilege runtime model.

9:00 PM - 10:00 PM AI on AWS / provider abstraction

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Spring AI provider integration -> OpenAI/Ollama/AWS Bedrock
- Spring AI abstraction

What to learn / focus on:

- Bedrock provider

- data residency
- provider abstraction
- model fallback

Skip / avoid in this block:

- Provider-specific code throughout business logic.

Hands-on work:

- Sketch config profile switching OpenAI/Bedrock/Ollama.

Interview questions you must answer:

- Why Bedrock?
- How avoid lock-in?

EXIT CRITERIA: Model provider abstraction plan.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 26 - BFS Level 3 - Full-stack production feature

Day goal: Build one feature from browser to database to deployment pipeline.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Full-stack coding drill

Assigned problems:

- LRU Cache
- Design Hit Counter

What to learn / focus on:

- data structure + API design crossover

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Backend feature slice

Course: Spring Boot 4, Spring 7 & Hibernate for Beginners - Chad Darby

Updated Jul 2026; Spring Boot 4 / Spring 7 / Hibernate 7

Course section / chapter to open:

- REST/JPA/security/testing sections as reference only

What to learn / focus on:

- DTO
- validation
- service rule
- repository
- transaction
- security
- tests
- event

Skip / avoid in this block:

- New course video unless blocked.

Hands-on work:

- Implement or fully specify one feature: GET /api/accounts/{id}/risk-summary or similar.

Interview questions you must answer:

- What belongs controller vs service?

EXIT CRITERIA: One production-style backend slice with tests.

1:00 PM - 3:00 PM System Design - feature architecture review**Course:** Active practice / mock - no required video before the first attempt**Course section / chapter to open:**

- Use design framework

What to learn / focus on:

- requirements
- API
- data
- cache/event
- failure
- scale
- security

Skip / avoid in this block:

- Over-design.

Hands-on work:

- Design same feature at 10x/100x usage and decide what actually changes.

Interview questions you must answer:

- What is bottleneck?

EXIT CRITERIA: Feature architecture that evolves incrementally.**3:00 PM - 5:00 PM Data/event implementation****Course:** Active practice / mock - no required video before the first attempt**Course section / chapter to open:**

- Postgres/Redis/Kafka references

What to learn / focus on:

- query/index
- cache
- event schema
- idempotency

Skip / avoid in this block:

- Adding all technologies regardless of need.

Hands-on work:

- Implement required index/cache/event only if justified.

Interview questions you must answer:

- Source of truth?

EXIT CRITERIA: Measured data path.**6:00 PM - 8:00 PM Frontend feature slice****Course:** React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- TypeScript + React relevant sections

What to learn / focus on:

- typed DTO
- hook
- component
- routing

- loading/error
- test

Skip / avoid in this block:

- any types.

Hands-on work:

- Build UI for backend feature and one test.

Interview questions you must answer:

- What if partial failure?

EXIT CRITERIA: End-to-end working feature.

8:00 PM - 9:00 PM CI/CD for feature

Course: GitHub Actions: The Complete Guide from Beginner to Expert - Lauro Fialho Muller

Updated Aug 2026; 20 sections / 115 lectures

Course section / chapter to open:

- workflow jobs / reusable workflows / security

What to learn / focus on:

- unit/integration/frontend test
- image build
- environment gate

Skip / avoid in this block:

- Manual undocumented deployment.

Hands-on work:

- Add/outline pipeline steps required for feature.

Interview questions you must answer:

- What gates prod?

EXIT CRITERIA: CI traceability from commit to deploy.

9:00 PM - 10:00 PM AI explanation layer

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- RAG / Tool Calling

What to learn / focus on:

- optional AI feature calls backend read tool and docs RAG

Skip / avoid in this block:

- AI write action.

Hands-on work:

- Add “Explain this risk summary” using RAG + current data tool.

Interview questions you must answer:

- Fresh data vs docs?

EXIT CRITERIA: End-to-end full-stack + AI feature.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).

- Commit or save the day's code/diagram/notes before stopping.

DAY 27 - BFS Level 3 - AI integration day

Day goal: Connect RAG, tools, MCP and safety to the existing Java platform.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: AI/backend coding drill

Assigned problems:

- Top K Frequent Elements redo
- Merge Intervals redo

What to learn / focus on:

- high-frequency interview patterns

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Spring AI architecture consolidation

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Spring AI - Say Hello
- RAG Unleashed
- Tool Calling
- Memory sections

What to learn / focus on:

- ChatClient
- advisors
- memory
- vector store
- tools
- structured output

Skip / avoid in this block:

- New basic AI lectures.

Hands-on work:

- Draw Spring beans/components for AI assistant.

Interview questions you must answer:

- What runs in request path?

EXIT CRITERIA: AI backend component diagram.

1:00 PM - 3:00 PM System Design - AI support platform**Course: Mastering the System Design Interview - Frank Kane / Sundog Education**

Updated Jul 2026; 9 sections / 73 lectures

Course section / chapter to open:

- Designing Generative AI Systems -> API/context, RAG, Agentic AI
- Design Interview Strategies

What to learn / focus on:

- model gateway
- RAG service
- vector index
- tool execution
- conversation store
- evaluation
- observability

Skip / avoid in this block:

- “Just call GPT API”.

Hands-on work:

- Design multi-tenant customer support AI with private data.

Interview questions you must answer:

- Tenant isolation?
- tool authorization?
- cost control?

EXIT CRITERIA: Interview-ready GenAI system design.**3:00 PM - 5:00 PM AI data pipeline****Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy**

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- RAG advanced sections
- MCP tools as context

What to learn / focus on:

- document ingestion
- versioning
- metadata
- embedding model upgrade
- deletion
- evaluation

Skip / avoid in this block:

- Never reindex/update stale docs.

Hands-on work:

- Design background ingestion job and reindex strategy.

Interview questions you must answer:

- How handle deleted docs?

EXIT CRITERIA: Production RAG data lifecycle.**6:00 PM - 8:00 PM AI frontend UX****Course: React - The Complete Guide (incl. Next.js, Redux) - Maximilian Schwarzmuller / Academind**

Updated Jan 2026; React 19; 40 sections / 727 lectures

Course section / chapter to open:

- HTTP/transitional states/forms
- TypeScript unions

What to learn / focus on:

- streaming response concept
- source citations UI
- tool activity indicator
- approval modal
- error/refusal

Skip / avoid in this block:

- Show confident prose without sources.

Hands-on work:

- Design assistant UI with answer + sources + “pending action approval”.

Interview questions you must answer:

- How display uncertainty?

EXIT CRITERIA: Trustworthy AI user experience.

8:00 PM - 9:00 PM MCP implementation

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- MCP -> Building an MCP Server using STDIO Part 1/2
- Building an MCP Server using remote invocation
- MCP Tool Filtering

What to learn / focus on:

- Spring MCP server
- tool annotations
- remote transport
- tool filtering
- logging

Skip / avoid in this block:

- Expose unrestricted admin tools.

Hands-on work:

- Expose read-only getPayment/getFraudAssessment tools; define auth context.

Interview questions you must answer:

- MCP vs direct @Tool?

EXIT CRITERIA: MCP server design for PAT-FOS.

9:00 PM - 10:00 PM AI evaluation + safety final

Course: From Java Dev to AI Engineer: Spring AI Fast Track - Eazy Bytes / Madan Reddy

Updated Jul 2026; Spring AI 2.x; 12 sections / 119 lectures

Course section / chapter to open:

- Evaluators
- Observability in Spring AI
- tool/MCP safety recap

What to learn / focus on:

- golden set
- trace

- token/latency
- PII
- prompt injection
- approval

Skip / avoid in this block:

- No regression tests.

Hands-on work:

- Run 10-case evaluation and record pass/fail.

Interview questions you must answer:

- What release gate for AI?

EXIT CRITERIA: AI quality gate ready for interview.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 28 - Interview Mode - Mock Day 1

Day goal: No passive course completion. Demonstrate skill under time pressure.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Mock Round 1

Assigned problems:

- 2 unseen problems, 45 min each

What to learn / focus on:

- full interview behavior

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Java + Spring Mock Interview

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course during first 90 minutes; patch with exact sections afterward

What to learn / focus on:

- Java collections/concurrency/JVM
- Spring IoC/transactions/security/JPA/testing/performance

Skip / avoid in this block:

- Notes.

Hands-on work:

- Run interviewer-style 15 questions; each answer 2-3 min + follow-up. Use last 30 min to patch top 2 misses.

Interview questions you must answer:

- Why? How? Failure? Trade-off?

EXIT CRITERIA: Score $\geq 7/10$ across core backend.

1:00 PM - 3:00 PM System Design Mock #1 - Payment / Order

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course first; use Design Strategies only for debrief

What to learn / focus on:

- full 45-min structure
- deep dive

- failure
- scale

Skip / avoid in this block:

- Memorized script.

Hands-on work:

- 45-min mock, 30-min challenge, 30-min redo, 15-min score.

Interview questions you must answer:

- Change requirement mid-interview.

EXIT CRITERIA: Complete design conversationally.

3:00 PM - 5:00 PM Database + Kafka Mock

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Patch using Postgres/Kafka/Redis/Mongo sections

What to learn / focus on:

- index/plan
- locking/MVCC
- Kafka duplicate/lag
- Redis cache
- outbox

Skip / avoid in this block:

- Definition-only.

Hands-on work:

- 10 questions + 3 failure scenarios.

Interview questions you must answer:

- Data too big? provider?

EXIT CRITERIA: Scorecard + misses.

6:00 PM - 8:00 PM React + TypeScript Mock

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Patch after

What to learn / focus on:

- hooks
- rendering
- typing
- auth
- state
- performance/testing

Skip / avoid in this block:

- Looking at code.

Hands-on work:

- Build a typed component from prompt and explain.

Interview questions you must answer:

- Why each hook?

EXIT CRITERIA: Frontend interview confidence $\geq 7/10$.

8:00 PM - 9:00 PM AWS/DevOps Mock**Course:** Active practice / mock - no required video before the first attempt**Course section / chapter to open:**

- Patch after

What to learn / focus on:

- Docker/K8s/CI/CD/AWS deployment

Skip / avoid in this block:

- Service buzzwords.

Hands-on work:

- Whiteboard local -> production.

Interview questions you must answer:

- Rollback/secrets/HA.

EXIT CRITERIA: 5-minute production deployment story.**9:00 PM - 10:00 PM AI Mock****Course:** Active practice / mock - no required video before the first attempt**Course section / chapter to open:**

- Patch after

What to learn / focus on:

- RAG/tools/MCP/memory/eval/safety

Skip / avoid in this block:

- Buzzwords.

Hands-on work:

- Explain AI assistant architecture and 3 failure cases.

Interview questions you must answer:

- How ground? authorize? evaluate?

EXIT CRITERIA: AI interview scorecard.**End-of-day checkpoint**

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 29 - Interview Mode - Unknown Problems

Day goal: Prove transfer learning rather than memorization.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Mock Round 2

Assigned problems:

- 2 unseen problems, 45 min each

What to learn / focus on:

- full interview behavior

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Backend Random Mock

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- random Java/Spring questions from RED/YELLOW list

Skip / avoid in this block:

- Prepared order.

Hands-on work:

- Use shuffled question list; answer with examples from PAT-FOS.

Interview questions you must answer:

- Follow-up until depth limit.

EXIT CRITERIA: Reduce all RED topics to YELLOW/GREEN.

1:00 PM - 3:00 PM System Design Mock #2 - Unknown

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No reference until end

What to learn / focus on:

- choose one: collaborative editor, feature flag service, ad click aggregator, distributed config service

Skip / avoid in this block:

- Known design template beyond framework.

Hands-on work:

- 45-min interview; interviewer changes scale/consistency halfway.

Interview questions you must answer:

- Why database? hot key? failure?

EXIT CRITERIA: Handle unknown design calmly.

3:00 PM - 5:00 PM Production Incident Mock

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Use notes after

What to learn / focus on:

- 5xx spike
- consumer lag
- DB pool
- Redis outage

Skip / avoid in this block:

- Jump to fix.

Hands-on work:

- Diagnose from synthetic metrics/logs.

Interview questions you must answer:

- What evidence?

EXIT CRITERIA: Structured incident response.

6:00 PM - 8:00 PM Full-stack feature mock

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- API + DB + React

Skip / avoid in this block:

- Copy PAT-FOS.

Hands-on work:

- Given new “transaction search” feature, design endpoint/query/index/UI and test.

Interview questions you must answer:

- Pagination? auth?

EXIT CRITERIA: End-to-end reasoning in 2 hours.

8:00 PM - 9:00 PM Cloud architecture random challenge

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- regional failure, cost, autoscale

Skip / avoid in this block:

- Only ideal architecture.

Hands-on work:

- Explain what you would simplify for 10k users vs 10M.

Interview questions you must answer:

- Cost trade-off?

EXIT CRITERIA: Right-size cloud design.

9:00 PM - 10:00 PM AI unknown scenario

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- HR policy RAG / coding assistant / support agent

Skip / avoid in this block:

- One-size-fits-all RAG.

Hands-on work:

- Choose architecture based on freshness, privacy and action needs.

Interview questions you must answer:

- RAG vs tool?

EXIT CRITERIA: Transfer AI patterns to new domain.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

DAY 30 - Final Simulation - No Notes

Day goal: Demonstrate the end-to-end engineer profile you want to present in interviews.

Fixed breaks: 12:00-1:00 PM and 5:00-6:00 PM. Study blocks total 12 hours.

8:00 AM - 10:00 AM LeetCode / DSA

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- Pattern: Final Coding Simulation

Assigned problems:

- 1 unseen medium + 1 follow-up

What to learn / focus on:

- no notes, production-quality explanation

Skip / avoid in this block:

- Do not watch solution before first timed attempt.
- Do not optimize before you can state a correct brute-force baseline.

Hands-on work:

- Solve in Python; speak the approach before coding.
- After each problem, write time and space complexity and one edge case.

Interview questions you must answer:

- Why is your chosen data structure appropriate?
- Can you derive the complexity instead of memorizing it?
- What changes if input is streaming or too large for memory?

EXIT CRITERIA: Finish the assigned problems or record exactly why one remains unsolved; maintain a mistake log.

10:00 AM - 12:00 PM Final Java/Spring Round

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- core Java
- concurrency
- Spring internals
- JPA/transactions
- security
- testing/performance

Skip / avoid in this block:

- New learning.

Hands-on work:

- Record or time yourself answering 12 final questions.

Interview questions you must answer:

- Answer with production examples and trade-offs.

EXIT CRITERIA: Final backend readiness score.

1:00 PM - 3:00 PM Final System Design Round

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- unknown prompt
- requirements
- scale
- API/data
- architecture
- failure
- trade-offs

Skip / avoid in this block:

- Prepared diagram.

Hands-on work:

- 45-min interview + 30-min challenges + 30-min clean redo + 15-min score.

Interview questions you must answer:

- Interviewer changes requirement.

EXIT CRITERIA: One interview-ready design from scratch.

3:00 PM - 5:00 PM Final Data/Distributed Systems Round

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- Postgres
- Redis
- Kafka
- Mongo
- outbox/saga
- consistency

Skip / avoid in this block:

- Definitions only.

Hands-on work:

- Answer 15 questions and two crash timelines.

Interview questions you must answer:

- Why/trade-off/failure.

EXIT CRITERIA: Final data readiness score.

6:00 PM - 8:00 PM Final Full-stack Round

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- React/TS + Spring API integration
- auth
- testing
- performance

Skip / avoid in this block:

- Copying existing implementation.

Hands-on work:

- Build/whiteboard one feature end-to-end and explain each layer.

Interview questions you must answer:

- Failure states?

EXIT CRITERIA: Final full-stack readiness score.

8:00 PM - 9:00 PM Final Cloud/DevOps Round

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- Docker/K8s/CI/CD/AWS/observability

Skip / avoid in this block:

- New cloud topic.

Hands-on work:

- Explain build -> deploy -> health -> rollback -> monitor.

Interview questions you must answer:

- Secrets/HA/cost.

EXIT CRITERIA: Final deployment story.

9:00 PM - 10:00 PM Final AI Round + 30-Day Debrief

Course: Active practice / mock - no required video before the first attempt

Course section / chapter to open:

- No course

What to learn / focus on:

- RAG
- tools
- MCP
- memory
- evaluation
- safety

Skip / avoid in this block:

- New AI topic.

Hands-on work:

- Explain AI feature, then score all tracks 1-10 and create next-30-day priority list.

Interview questions you must answer:

- What still needs depth?

EXIT CRITERIA: Final scorecard + next sprint.

End-of-day checkpoint

- Write 3 things you can now explain without notes.
- Write 2 mistakes or weak areas to revisit in the next BFS pass.
- Score each track touched today from 1-5 (1-2 RED, 3 YELLOW, 4-5 GREEN).
- Commit or save the day's code/diagram/notes before stopping.

Appendix A - Final 30-Day Readiness Scorecard

Track	Target capability	Score /10	Next action
LeetCode / DSA	Solve unseen medium in 35-45 min, test, complexity, follow-up.		
Java	Collections, concurrency, JVM, modern Java, performance.		
Spring Boot	IoC, MVC, JPA, transactions, security, testing, performance.		
System Design	Requirements -> scale -> API/data -> architecture -> deep dive -> failures/trade-offs.		
PostgreSQL	Schema, indexes, EXPLAIN, isolation/MVCC, locks, partitioning.		
Kafka	Partitions, groups, ordering, offsets, lag, retries/DLQ, idempotency, outbox.		
Redis	Cache-aside, invalidation, data structures, rate limiting, coordination trade-offs.		
MongoDB	Document modeling, indexes, aggregation, replica/shard concepts.		
React/TypeScript	Hooks, state ownership, typing, routing/auth, performance, testing.		
Docker/Kubernetes	Images/layers/Compose, deployments/services/config/probes/scaling.		
CI/CD	Pipeline-as-code, parallel tests, artifacts, security, deployment/rollback.		
AWS	ALB/ECS/RDS/ElastiCache/S3/CloudFront/messaging/monitoring/security.		
AI	Spring AI, embeddings, RAG, tools, memory, MCP, evaluation, safety.		

Appendix B - Daily Study Contract

- No passive video binge: pause and predict before the instructor explains the answer.
- Every new technology must be tied to a requirement, failure mode, or trade-off.
- For interviews, use this answer pattern: What it is -> Why I use it -> How it works -> Failure/trade-off -> Example from PAT-FOS.
- LeetCode: always state brute force, improved approach, time/space, and one follow-up.
- System design: do not start with technologies. Start with requirements and scale.
- Use PAT-FOS as proof: point to idempotency, Kafka, Redis, Postgres, Mongo, locking, security, CI/CD and React when relevant.